

Fire DNF, Fire HSEF - electric

Installation and operating instructions



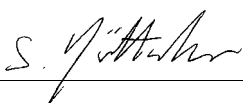
Declaration of conformity

GB: EC declaration of conformity

We, Grundfos, declare under our sole responsibility that the products Fire DNS and Fire HSEF, to which this declaration relates, are in conformity with these Council Directives on the approximation of the laws of the EC Member States:

- Machinery Directive (2006/42/EC).
Standards used: EN 12100-1:2003, EN 12100-2:2003, EN 809:1998.
- Low Voltage Directive (2006/95/EC).
Standards used: EN 60204-1:2006, EN 60439-1:1999.
- EMC Directive (2004/108/EC).
Standards used: EN 61000-6-2:2005, EN 61000-6-3:2007.

Wahlstedt, 15th April 2010



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Person authorised to compile technical file and
empowered to sign the EC declaration of conformity.

Original installation and operating instructions.

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**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document



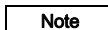
Warning

If these safety instructions are not observed, it may result in personal injury.



Caution

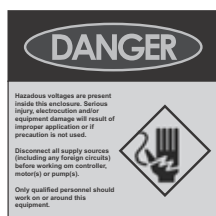
If these safety instructions are not observed, it may result in malfunction or damage to the equipment.



Note

Notes or instructions that make the job easier and ensure safe operation.

Safety instructions placed directly on the fire pump set:

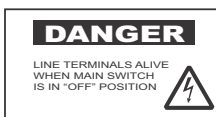


Warning

Hazardous voltages are present in this enclosure. Improper use or failure to take precautions can cause serious injury, electric shock and/or damage to the equipment.

Disconnect all supply sources (including any foreign circuits) before working on controller, motor or pump.

Only qualified personnel should work on or around this equipment.



Warning

The mains terminals are still live when the main switch is in position "OFF".



Warning

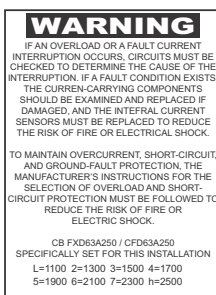
Phase-sensitive equipment. Connect in ABC order.



Warning

The enclosure and withstand rating are applicable only if all door locks are in closed position.

The ambient temperature must be 5 to 50 °C.



Warning

If an overload or a fault current interruption occurs, circuits must be checked to determine the cause of the interruption. If a fault condition exists, the current-carrying components should be examined and replaced if damaged, and the integral current sensors must be replaced to reduce the risk of fire or electric shock.

To maintain overcurrent, short-circuit and earth-fault protection and to reduce the risk of fire or electric shock, the manufacturer's instructions for selection of overload and short-circuit protection must be followed.



Lifting points for the entire fire pump set.

2. Applications

The Grundfos Fire HSEF and DNF pump sets are designed for fire systems for supplying water to hose reels, fire hydrants and sprinkler systems.



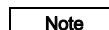
Warning

The fire pump set must only be used for the applications mentioned. Any other use is considered improper use. Grundfos cannot be held responsible for damage caused by improper use. The liability rests solely with the operator.

The Grundfos Fire HSEF and DNF pump sets with electric motor must not be used for ordinary pumping of liquids or common pressure boosting in daily work. They should only be used for firefighting.

The pump controller must not be used as a junction box for supplying other equipment.

3. Delivery, transport, storage



Note

Check the fire pump set carefully on delivery against the papers and inspect for damage incurred during transport. Immediately notify the carrier of any damage or shortage found. Transport and store it correctly before installation.

3.1 Delivery

The fire pump set is delivered from factory in an open wooden box or wooden/cardboard box which is specifically designed for transport by fork-lift truck or similar device.

3.2 Transport

For lifting the entire fire pump set, use the lifting holes of the base frame. When lifting, the lifting point should always be above the centre of gravity of the fire pump set. See fig. 1.



Warning

Lifting the fire pump set should only be done by qualified personnel.

When lifting the entire fire pump set, never use the lifting eyes of the individual components.

Use only suitable lifting equipment in proper condition. See also weight specifications in section 7. Technical data.

During transport of the fire pump set, the cabinet door must be closed and locked.

Do not stand or place hands under a component in such a way that the component would fall on the hands if dropped.

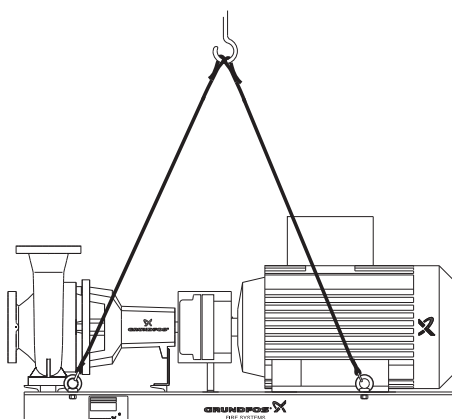


Fig. 1 Correct lifting of fire pump set

3.3 Storage

To prevent water, dust, etc. from entering the pump, all openings must be covered until pipes are connected. Apply a suitable antirust agent on all machined, non-coated surfaces.

The antirust agent must meet these requirements:

- It must not attack rubber parts.
- It must be easy to remove.
- It must be applied according to the instructions of the manufacturer.

If the pump must be stored for more than six months before start-up, it must be protected against corrosion and damage:

1. Remove the stuffing box packing.
2. Spray the interior part of the pump case and the stuffing box with a water-soluble type of antirust agent.
3. If the pump has not been connected to the pipework, cover the pump suction and discharge flanges with full natural rubber gasket material, and blank off these openings with metal blanking flanges. Use at least four bolts for fastening.
4. Cover the opening of the stuffing box with a non-hygroscopic tape. The stuffing box gland may be left on the pump shaft, but must be wired or otherwise securely fastened in position.
5. Coat all exposed painted surfaces with an antirust agent that can be readily removed with a petroleum distillate product.
6. Make sure that all exposed painted surfaces are dry, clean and free of grease and other contaminates.
7. Cover the pump with a weather-resistant cover of waterproof paper or plastic material to protect it against dirt and dust.
8. Inspect the pump at regular intervals during storage.
9. Rotate the shaft by hand every four to six weeks to prevent pitting of the bearings.

4. Product description

4.1 Fire pump set

The fire pump set consists of a pump, an electric motor and a controller. Pump and motor are connected via a flexible coupling. All components are adapted to each other and mounted on a common base frame ready for installation. The base frame is made of steel channel. The controller can also be placed separately. The fire pump set comes ready for operation.

As an example, the most important components of the fire pump set are shown in fig. 2. It shows an HSEF fire pump set with a Grundfos horizontal split case pump and an electric motor, type MMG. Pump and motor are connected by a flexible coupling. The controller is installed separately.

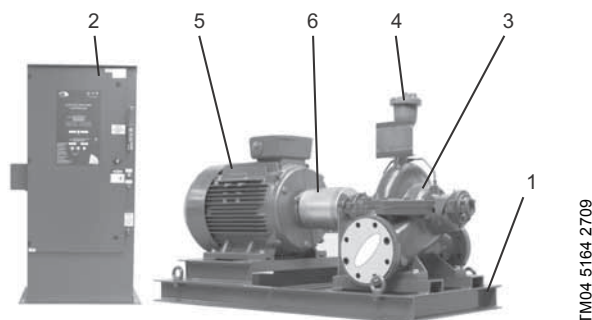


Fig. 2 Example of fire pump set

Pos.	Component
1	Base frame
2	Controller
3	Pump
4	Automatic air release valve
5	Motor
6	Coupling with coupling guard

4.2 Pump

The fire pump set comes with a Grundfos HSEF horizontal split case pump or a Grundfos DNF end-suction pump. The pumps are FM-approved and UL-listed.

The impeller diameter of both pumps can be reduced to customise the pump performance to a certain duty point. The actual impeller diameter may therefore deviate from the standard diameters stated in catalogues, data sheets etc. The actual impeller diameter is stated on the pump nameplate.

4.2.1 Split case pump

The Grundfos HSEF horizontal split case pump is a non-self-priming, single-stage pump. It has inline axial suction and discharge ports with PN 10 or PN 16 ANSI flanges. An adapter flange from ANSI to DIN is available and must be ordered separately. The pumps are equipped with a stuffing box, type SNEA or SNFA (see section 6.1.3 *Type key for stuffing box*), and with an automatic air release valve making venting unnecessary.

4.2.2 End-suction pump

The pumps are non-self-priming, single-stage Grundfos DNF standard pumps with spiral housing. The pumps have an axial suction port and a radial discharge port with PN 10 or PN 16 DIN flanges.

The DNF pump is equipped with a stuffing box, type SNEA or SNFA. See section 6.1.3 *Type key for stuffing box*.

4.3 Motor

The pumps are driven by a 2- or 4-pole Grundfos MMG-E motor (15-315 kW) or a Siemens motor (355 kW). All motors are produced according to IEC 60034. The rated motor power is adapted to the power requirement of the pump. The combinations of motors and pumps are shown in section 19. *Dimensions, weights and motor data*.

4.4 Controller

4.4.1 General description

The fire pump set is controlled via a Tornatech controller especially sized for electric motors driving fire pumps. The controller starts the pump automatically and monitors the motor. It is FM-approved.

As soon as the sprinklers are activated due to an alarm and water is consumed, the pressure in the discharge pipe will be reduced. If the pressure becomes lower than the cut-in pressure set on the control panel, the pump will start automatically.

Normally, the pump is stopped manually via the controller. But automatic stop is also possible if all starting causes have returned to normal.

For test run and set-up, manual operation is possible.

The individual controller models differ mainly by starting method and power range.

Starting method, model GPA

This model is used where the electricity works or the capacity of the power source permits direct-on-line starting.

Full voltage is applied to the motor as soon as the controller receives a start command.

Starting method, model GPY

When the controller receives a start command, the motor is connected in star. After a time delay, it changes over to delta connection with full voltage to the motor windings. This controller is of the open transition type. The motor is disconnected from the mains during the transition from start (star) to run (delta) mode.

4.4.2 Controller design

The controller is placed in a control cabinet mounted on the base frame (compact unit) or delivered separately for wall- or floor-mounting (flex unit). The controller is operated by means of the control panel in the cabinet door. See fig. 3. The control cabinet supplies the motor with the electricity required for starting and operating the motor.

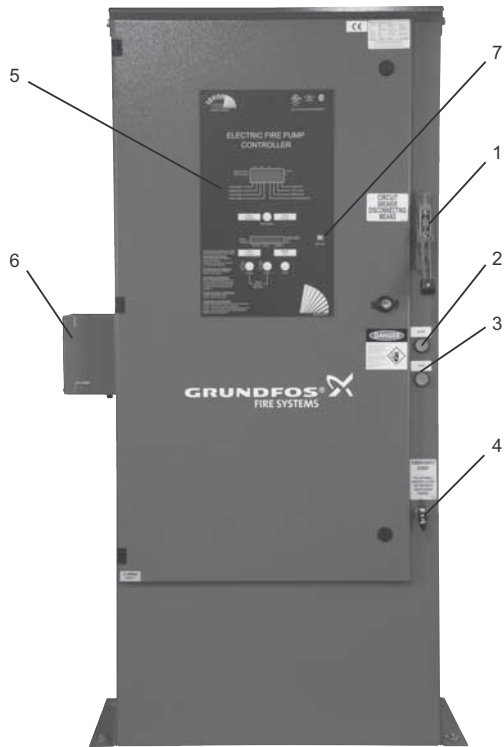


Fig. 3 Control cabinet

Pos.	Operating element
1	Circuit breaker (A1)
2	Start button (A2)
3	Stop button (A3)
4	Emergency start handle (A4)
5	Control panel
6	Box with pressure sensor
7	USB port

The design of the control panel is shown in fig. 4. It consists of two displays, four indicator lights and four buttons. Display D1 shows the status of the power supply. Display D2 shows the status of pressure data and allows for setting of pressure values. See section 5.2 *Settings via display*.

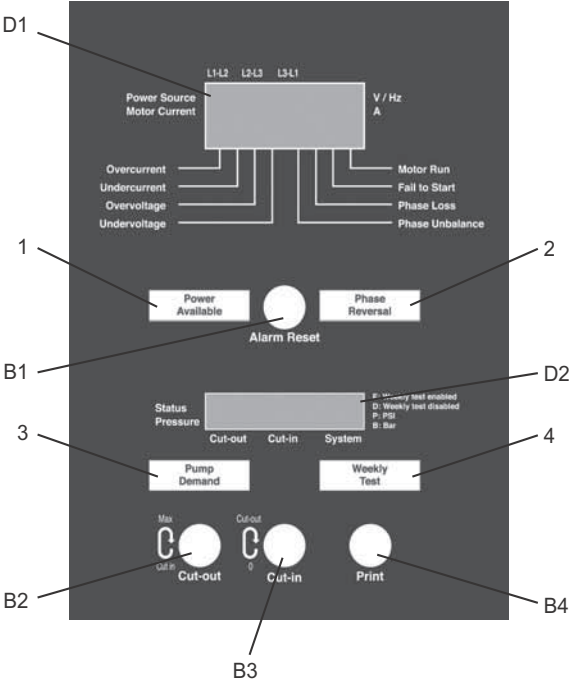


Fig. 4 Control panel

The controller has two DIP switches for settings. See section 5.1 *Settings via DIP switches*. The internal components of the controller are shown in fig. 5 for model GPA and in fig. 6 for model GPY.

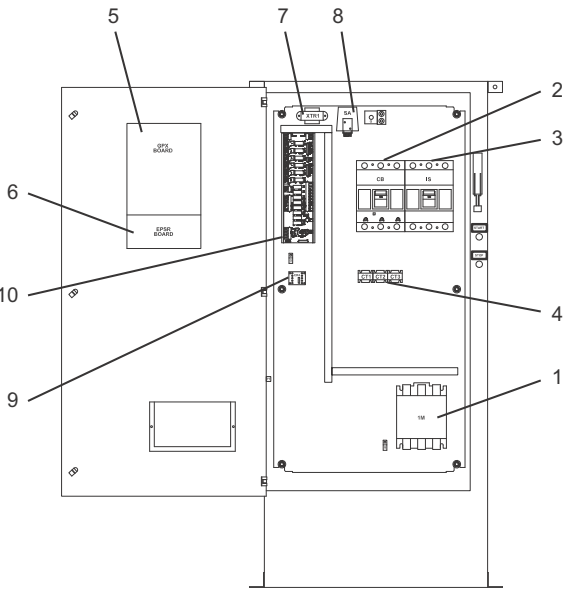


Fig. 5 Internal components of model GPA

Pos.	Designation	Description
1	1M	Main contactor
2	CB	Circuit breaker
3	IS	Isolating switch
4	CT1 - CT3	Measuring devices
5	GPX board	-
6	Pressure board (EPSR)	-
7	XTR1	Transformer
8	SA	-
9	CR4	Control relay
10	-	Relay board

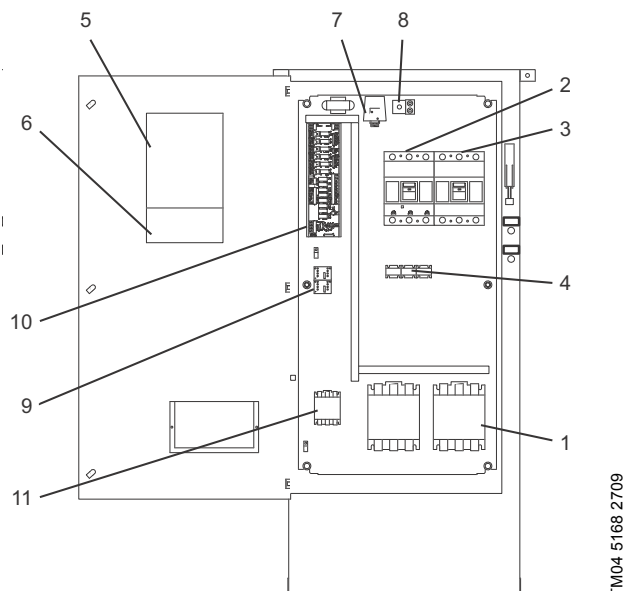


Fig. 6 Internal components of model GPY

Pos.	Designation	Description
1	1M, 2M	Main contactor
2	CB	Circuit breaker
3	IS	Isolating switch
4	CT1 - CT3	Measuring devices
5	GPX board	-
6	Pressure board (EPSR)	-
7	XTR1	Transformer
8	SA	-
9	CR4, CR5	Control relay
10	-	Relay board
11	1S	-

The relay board has six standard alarm relays and 13 optional alarm relays for indication of operation and alarm status in a building management system or similar.

4.4.3 Operating elements

The operating elements are placed on the cabinet door and control panel. See fig. 3.

Operating elements on the cabinet door

Circuit breaker A1 (fig. 3, pos. 1)

The circuit breaker interrupts the power supply from the mains.

[START] A2 (fig. 3, pos. 2)

This button can be used to start the pump regardless of system pressure.

[STOP] A3 (fig. 3, pos. 3)

The button is used for manual stop.

Emergency start handle A4 (fig. 3, pos. 4)

For starting the pump in case of failure, see section 12.2.2 *Manual operation with emergency start handle*.

Operating elements on the control panel

[Alarm Reset] B1 (fig. 4)

This button is used to reset fault indications. When the button is pressed for more than five seconds, display D1 (fig. 4) shows input/output status. When pressed for more than ten seconds, a ten second phase-reversal simulation is carried out.

[Cut-out] B2 (fig. 4)

This button is used to set the pressure limit for stopping the pump. It is enabled with a DIP switch. See section 5.1 *Settings via DIP switches*.

It is also used when setting the controller. See 5.2 *Settings via display*.

[Cut-in] B3 (fig. 4)

This button is used to set the pressure limit for starting the pump. It is enabled with a DIP switch. See section 5.1 *Settings via DIP switches*.

It is also used when setting the controller. See 5.2 *Settings via display*.

[Print] B4 (fig. 4)

If the controller is equipped with an optional printer, this button is used to print a resume of events of the last 15 days and pressure data of the last seven days.

4.4.4 Display D1 (power status)

In normal mode, the display shows the status of the fire pump set and alarm messages.

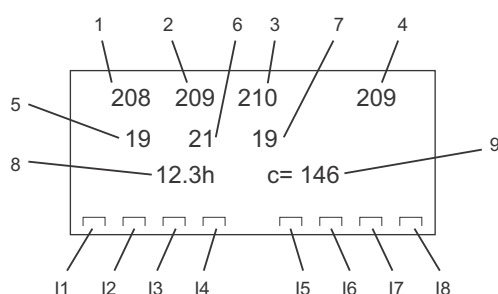


Fig. 7 Display 1 in normal mode

The display is divided into four lines. The first three lines show these data:

Pos.	Status
1	Incoming voltage on phase 1
2	Incoming voltage on phase 2
3	Incoming voltage on phase 3
4	Frequency of the mains
5	Current on phase 1
6	Current on phase 2
7	Current on phase 3
8	Operating hours
9	Number of pump starts

In case of alarms, the third line shows alarm messages instead of operating hours and number of pump starts.

The fourth line shows seven indicator lights. The table below shows when the indicator lights will flash and how to reset them.

Additionally, indicator I8 shows that the pump is running.

Pos.	Indication	Description	Reset
I1	Overcurrent	$I > 130\%$ of I_N for more than 20 seconds.	M
I2	Undercurrent	$I < 30\%$ of I_N for more than 20 seconds.	M
I3	Overvoltage	$U > 110\%$ of U_N for more than 1 second.	M
I4	Undervoltage	$U < 85\%$ of U_N for more than 1 second.	M
I5	Phase Unbalance	Over 3 % for more than 1 second.	M
I6	Phase Loss	Power missing for more than 1 second.	A
I7	Fail to Start	$I < 5\%$ for more than 20 seconds.	M
I8	Motor Run	Mains contactor is closed.	A

M: Manual reset with [Alarm Reset] button B1.

A: Automatic reset.

Status mode

The display can be switched to status mode by pressing [Alarm Reset] button B1 (fig. 4) for 5 seconds.

The first three lines shows the status of inputs and outputs. The fourth line shows the maximum voltage (OV) applied to the controller and the earth fault limit (GND). The earth fault detection is optional (option B13). The display switches back to normal mode after 10 minutes, or when the [Alarm Reset] button is pressed.

12345678
Input: 00001000
Output: 10000000

Fig. 8 Display 1 in status mode

The table below gives an overview of inputs and outputs.

Pos.	Input	Output
1	Alarm 1	Power Available
2	Alarm 2	Phase Reversal
3	Alarm 3	Pump Room Relay
4	Alarm 4	Motor Alarm Relay
5	Motor run	Lock Rotor (shunt trip)
6	Alarm 5	Alarm A
7	DIP ABC-CBA	Alarm B
8	Reset PB	Alarm C

Simulation mode

The display mode can be switched to phase-reversal simulation by pressing [Alarm Reset] button B1 (fig. 4) for more than 10 seconds. See fig. 8.

During this simulation, indicator light "Phase Reversal" (fig. 4, pos. 2) is on, and alarm relay "Ph. Rev." is activated.

Phase Reversal Simulation

Fig. 9 Display 1 in simulation mode

4.4.5 Display D2 (pressure status)

The display shows the pressure status of the system.

We 23:23:32	E
245 252 262	P

Fig. 10 Display 2

The display is divided into two lines. The first line shows day and time, or status messages. The day and time are always shown. If a message must be shown, the display will be scrolling successively at intervals of ± 1 second.

The last digit indicates whether the weekly test is enabled (E) or disabled (D). If a timer count-down (on timer and off timer) is activated, the remaining time will be shown to the right instead of E or D.

We 23:23:32	22
246 252 262	P

If manual stop is required (automatic stop function is not enabled), the word "MANU" will be shown to the right.

We 23:23:32	MANU
245 252 262	P

The second line shows the pressure status. The first group of digits shows the cut-out pressure, the second group shows the cut-in pressure, and the third group shows the actual system pressure. The pressure can be displayed in PSI (P) or bar (B).

The display also gives access to the setting menus of the controller. See section 5.2 *Settings via display*.

4.4.6 Indicator lights on control panel

Four indicator lights show the status of the controller and pump. The indicator lights and their indication are described in the table below.

Pos.	Indicator light	Indication
1	Power Available	This indicator light indicates that the power is acceptable on all three phases, and the phase sequence is correct. The power is acceptable if the voltage is more than 85 % of rated voltage.
2	Phase Reversal	This indicator light indicates phase reversal of the power source.
3	Pump Demand	This indicator light indicates an automatic start condition. It is activated when the system pressure falls below the cut-in pressure. It is deactivated when the run time has expired (if the function is enabled), or when [STOP] button A3 (fig. 3, pos. 3) is pressed.
4	Weekly Test	This indicator light indicates that the system is in operation due to a weekly test. The weekly test is programmable. The weekly light is also on during manual test.

4.5 Functions

Control and monitoring take place via the functions described in the following subsections.

4.5.1 Operating functions

Automatic operation

The controller starts the pump when the pressure sensor detects that the system pressure is lower than the cut-in pressure.

The cut-in and cut-out pressures can be set by means of buttons [Cut-in] B3 and [Cut-out] B2 (fig. 4). See section 11. *Start-up*.

If the pump does not start, indicator light "Fail to Start" will be on (fig. 5, pos. 17).

The pump can be stopped with [STOP] button A2 (fig. 3, pos. 3) when the system pressure is higher than the cut-out pressure.

Otherwise, indicator light "Pump Demand" is on, and the pump can only be stopped by turning the circuit breaker to "OFF".

Automatic stop is also possible. See below.

Automatic stop

In automatic operation, the pump stops automatically when all starting causes have returned to normal. To use automatic stop, follow the procedure in section 5.2 *Settings via display*, subsection *Menu 3 (timers)*. Normally, the pump is stopped after 10 minutes, but the run time can be changed. See section 5.2 *Settings via display*.

Manual operation

For test runs and start-up, manual operation is possible by means of [START] button A2 (fig. 3, pos. 2) regardless of system pressure. See section 12.2.1 *Manual operation with buttons [START] and [STOP]*.

To stop the pump, press [STOP] button A3 (fig. 3, pos. 3).

Emergency operation

The pump can be started manually with emergency start handle A4 (fig. 3, pos. 4). The handle can be kept in a closed position. See section 12.2.2 *Manual operation with emergency start handle*.

Remote control

The pump can be started by the momentary opening of a remote contact, independent of the pressure sensor.

Start via an external signal (deluge valve/flowmeter)

The controller can be started by opening a normally closed contact on the fire protection equipment (deluge valve), independent of the pressure sensor.

If option A4 is supplied (flowmeter), the pump can be started by closing a normally open contact from the flowmeter, independent of the pressure sensor. If option C17 was chosen, control relay CR39 will be activated when the contact of the flowmeter is open. If the pump has started because of the flow switch, and the controller did not detect a pressure drop, the pump will stop when the flow switch contact opens. If the controller is set for manual stop only, press [STOP] button A3 (fig. 3, pos. 3) to stop the pump. When the water flow has subsided, the pump will. If the controller is set for automatic stop, the pump will stop when the water flow has subsided, and the run time has expired.

Automatic weekly test

The controller can be set for a weekly test. See section 5.2 *Settings via display*. When the weekly test is enabled, the pump starts at the set start time and stops at the set stop time or after the run time set for automatic stop, whichever is the shorter. The pump can be stopped with [STOP] button A3 (fig. 3, pos. 3) before the test is finished (provided that the system pressure is higher than the cut-out pressure). If the pump was started due to an automatic weekly test, indicator light "Weekly Test" (fig. 4, pos. 4) will be on.

If option C4 was chosen, alarm relay "Weekly T." will be activated.

Manual test run

The pump can be started for a running period of 10 minutes by pressing buttons [Cut-in] B3 and [Cut-out] B2 (fig. 4) at the same time. Pressing these buttons simulates a pressure drop in the system.

Start delay

In systems with several pumps, it may be necessary to set a start delay to prevent all motors from starting at the same time.

The start delay is selectable. See section 5.2 *Settings via display*, subsection *Menu 3 (timers)*.

Phase-reversal test

A phase reversal can be simulated by pressing the [Alarm Reset] button for more than 10 seconds. Indicator light "Phase Reversal" (fig. 4, pos. 2) lights up, and alarm relay "Ph. Rev." is activated.

Lockout by other equipment (option A6)

When the NO contact is open, the controller is enabled. If the NO contact is closed, the controller will be locked and will not start the pump in case of pressure drop (automatic start). The pump can always be started manually with the [START] button.

Interlock of other equipment (option A7)

The contacts of relay CR 34 are activated in order to prevent other equipment from starting.

Division into zones (options A9, A10, A11)

Controllers of different zones can be interconnected so that the equipment for the different zones can be started in a certain order.

Anti-condensation heater (options D14, D14A, D14B)

To avoid forming of condensation, the control cabinet can be equipped with an anti-condensation heater with thermostat (option D14) or humidistat (option D14A), or both (option D14B).

Motor heater (option D27)

To avoid condensation in the terminal box and the stator, the controller has a contact for connecting an external single-phase motor heater, if option D27 has been selected.

4.5.2 Monitoring functions

Monitoring of motor operation

If the power supply has been connected correctly, indicator light "Power Available" (fig. 4, pos. 1) will be on, and alarm relay "EPSR Ready" will be activated to signal that the pressure board (EPSR) is ready and functioning. If option C16 was chosen, control relay CR27 will also be activated to indicate that the control voltage is okay.

If the pressure in the discharge pipe falls below the cut-in pressure, indicator light "Pump Demand" (fig. 4, pos. 3) will be on. When the pump is running, indicator I8 (fig. 7) is on, and contact "Run NC/NO" is activated. If option C4 has been ordered, the extra motor run contact will also be activated. If option C14 has been ordered, control relay CR25 will also be activated.

If the pump remains below 5 % for more than 20 seconds so that the pump does not start, indicator light I7 "Fail to Start" in display D1 will be on. If option C15 has been ordered, control relay CR26 will be activated.

In case of an overcurrent or undercurrent condition, alarm relays "Motor Tr." (fig. 5 and 6, pos. 9) will be activated, and the corresponding indicator light I1 or I2 in display D1 (fig. 7) will light up.

Monitoring of power supply

Alarm relays "Pow. Av." (fig. 5 and 6, pos. 9) will be disabled in case of signal loss of any phase or power, and indicator light "Power Available" (fig. 4, pos. 1) will be off. Indicator I6 in display D1 (fig. 7) will light up.

In case of overvoltage, undervoltage or phase failure, the corresponding indicator lights I3, I4 and I5 in display D1 (fig. 7) will be on, and alarm relays "Room Al." (fig. 5 and 6, pos. 9) will be activated.

If an earth fault is detected, the optional control relay CR24 will be activated.

Monitoring of inlet pressure (option C5)

If a pressure switch is installed in the suction pipe, the optional control relay CR18 will be activated in case of a pressure drop.

Monitoring of discharge pressure (option C6)

If a pressure switch is installed in the discharge pipe, the optional control relay CR19 will be activated in case of a pressure drop.

Monitoring of ambient temperature (option C7)

If a thermostat for measuring the room temperature is installed, the optional control relay CR20 will be activated when the temperature falls below a set temperature.

Monitoring of water reservoir level (option C10/C18)

The level in the water reservoir is monitored by two float switches (low level and high level). If a low level is detected, the optional control relay CR21 will be activated, provided option C10 has been ordered.

If a high level is detected, the optional control relay CR40 will be activated, provided option C18 has been ordered.

Monitoring of motor (option C11/C12)

If a motor temperature sensor is connected to terminals 89-90 via a shielded cable, the optional control relay CR22 will be activated in case of a high motor temperature.

If a vibration sensor is installed, the optional alarm relay CR23 will be activated in case of too strong vibrations indicating a motor or pump fault.

Display texts

Most operating and alarm conditions can be shown as digital texts in display D1.

Alarm panel (option B11b) and alarm bell (option D24)

The alarm panel provides an audible alarm and silence button for motor run, phase reversal and phase failure.

An audible alarm for any optional digital text indication or alarm relay is available as option. The alarm bell on the side of the controller is also sounding when the isolating switch is in position "OFF". During start-up, the audible alarm can be silenced for a period of 15 minutes by pressing the silence button.

Recording of events and pressure data

The controller records events of the last 15 days and pressure data of the last seven days. The information is accessible either by using the USB port (standard, see section 12.4.1 *Retrieving data via the USB port*), or the printer (option D17A, see section 12.4.2 *Printing of data*). A modem can be installed for remote monitoring (option D25).

5. Settings

Settings are made via the two DIP switches and the menu in display D2.

5.1 Settings via DIP switches

The following settings can be made via the two DIP switches:

- phase-reversal reference
- locking or unlocking of buttons [Cut-in] B3 and [Cut-out] B2 for setting the cut-in and cut-out pressures.

The two DIP switches are located on the back of the cabinet door. The configuration and factory settings are described below.

Setting of upper DIP switch

Inp 1			Do not change
Inp 2			Do not change
Inp 3			Do not change
Inp 4			Do not change
Ph mon			Do not change
ABC-CBA			Phase-reversal reference

Phase-reversal reference (switch No 6)

If indicator light [Phase Reversal] is on, change the DIP switch setting "ABC-CBA" to another setting using a pencil.

Setting of lower DIP switch

Lock-Unlock			Pressure setting
Norm-Exp			Do not change
mV - 0..5V			Do not change
cw/wo valve			Do not change
Not used			Do not change
Not used			Do not change

Unlocking or locking of buttons for setting of cut-in and cut-out pressures (switch No 1)

When DIP switch "Lock-Unlock" is in position "ON", the buttons [Cut-in] B3 and [Cut-out] B2 will be enabled. When DIP switch "Lock-Unlock" is in position "OFF", setting of cut-in and cut-out pressures is not possible.

5.2 Settings via display

13 menus give access to setting of several parameters:

- time and date, and time for automatic weekly test (menu 1)
- pressure unit (menu 2)
- timers (menu 3)
- pressure differential (menu 4)
- calibration of pressure sensor (menu 5)
- erase memory (menu 6)
- maximum pressure (menu 7)
- low system pressure (menu 8)
- high system pressure (menu 9)
- print (menu 10)
- pressure log (menu 11)
- event log (menu 12)
- status of inputs and outputs (menu 13).

The menus are listed in the order appearing in display D2.

Menus 4 to 9 are reserved for service staff and will be described in the service instructions.

General procedure

- Press [Cut-in] button B3 (fig. 4) for more than 5 seconds to get access to the setting menus. Keep the button pressed.
- To go to the next menu, press [Cut-out] button B2 (fig. 4) repeatedly until the desired menu appears. Then release the [Cut-in] button.
- To select a value, press [Cut-in] button B3.
- To set a value, press [Cut-out] button B2.
- To save the setting and return to the "normal" display, press [Cut-out] button B2.
- If no button is touched for 30 seconds, the setting will not be saved, and the display will return to normal mode.

Menu 1 (time and date, and time for automatic weekly test)

In the first line, the user can set time and date.

In the second line, the user can enable or disable a weekly test: ("WT : ON") / ("WT : OFF").

22:36 02 - Ja - 2006
WT : ON NEXT

If a weekly test is enabled, "NEXT" will open a second window.

STOP START Mo
05:35 05:30 SAVE

In the first line, the day of week for the test is set. In the second line, the stop and start times are set.

Menu 2 (pressure unit)

In this menu, the user can select the unit of the displayed pressure. The pressure can be displayed in bar or psi.

Unit : PSI
SAVE

TM04 5150 2709

TM04 5151 2709

TM04 5152 2709

Menu 3 (timers)

In this menu, two different timers can be set.

OFF Tim: 600 Auto
ON Tim: 10 SAVE

TM04 5153 2709

a) Run time before automatic stop

This function is set in line 1. It is possible to set manual stop ("MANU") or automatic stop ("AUTO"). The delay is set in seconds (0 to 999).

If manual stop is selected, the pump can only be stopped with the [STOP] button. The timer value has no impact in this mode.

If automatic stop is selected, the pump will be stopped automatically after the time set.

The timer starts counting when the system pressure is higher than the cut-out pressure.

If the system pressure falls below the cut-out pressure, the timer will be reset automatically to its initial value.

Note

The remaining time before stop is displayed in the top right of the display.

b) Start delay

A start delay can be set in seconds (0 to 99) in line 2. The pump starts after the delay. The timer starts counting when the system pressure falls below the cut-in pressure. It is reset to its initial value when the system pressure reaches the cut-out pressure.

When start delay has been set, the remaining time before start is displayed in the top right of the display.

Note

Menu 10 (print)

In this menu, the user can select how pressure data are to be shown when printed (if a printer is installed). The print mode can be set as "TEXT" or "GRAPH".

"TEXT": Each pressure record is printed on a separate line.

Print Mode : TEXT
SAVE

TM04 5154 2709

"GRAPH": The pressure record is printed as a bar graph.

Print Mode : GRAPH
NEXT

TM04 5155 2709

When "GRAPH" is selected, a second window can be opened to select the axis value. The Y axis is the pressure axis.

Y max : 300
Y min : 200 NEXT

TM04 5156 2709

The X axis is the time axis. The selection is 1 h or 1/4 h. The grid can be set between 0 and 4. If set, the grid line will be drawn to help reading the pressure.

X : 15 min.
Grid : 2 SAVE

TM04 5157 2709

Menu 11 (pressure log)

In this menu, the user can view the pressure log.

Pressure Log

TM04 5158 2709

The first line shows date/month and time, the second line shows the pressure.

03 - mar 12 : 36 : 18
125

TM04 5159 2709

Press [Cut-in] button B3 to move to the next log or [Cut-out] button B2 to move to the previous log.

To exit the menu, do not press any button for more than 30 seconds, or press [Cut-in] button B3 for more than 10 seconds.

Menu 12 (event log)

In this menu, the user can view the event log.

Event Log

TM04 5160 2709

The first line shows date/month and time, the second line shows the event.

03 - mar 12 : 27 : 03
Pump start

TM04 5161 2709

Press [Cut-in] button B3 to move to the next log or [Cut-out] button B2 to move to the previous log.

To exit the menu, do not press any button for 30 seconds, or press [Cut-in] button B3 for more than ten seconds.

Menu 13 (status of inputs and outputs)

In this menu, the user can view the status of inputs and outputs.

In: 1.3.4.5 Out: 6.8
A: 2A5Ah B: 3684h

TM04 5162 2709

The first line shows the numbers of activated inputs and outputs.

Inputs:

- 1: Pump room alarm
- 2: Motor fault
- 3: Motor run
- 4: Stop button
- 5: DIP switch (with without solenoid valve).

Outputs:

- 6: Weekly test
- 7: Pressure board (EPSR) ready
- 8: Pump demand.

The second line shows the hexadecimal value of analog input 1 (0 to 5 V) and input 2 (0 to 100 mV).

6. Identification

The fire pump set can be identified by the type designation on the nameplate. See fig. 8.

6.1 Type keys

An explanation of the type designation is given in the following subsections.

6.1.1 Type key for fire pump set

Example DNF:	Fire	DN	F	80	-25	/260	E	A	X	A	A	B	B
Example HSEF:	Fire	HSE	F	8	-15	/323	E	B	X	D	A	B	C
Fire: Fire system													
Pump type													
DN: End-suction pump													
HSE: Horizontal split case pump													
F: Pump is approved for firefighting													
Nominal diameter of discharge port													
[mm] for DN													
[inch] for HSE													
Pump housing size													
[cm] for DN													
[inch] for HSE													
Actual impeller diameter [mm]													
Driver													
D: Diesel engine													
E: Electric motor, 50 Hz													
F: Electric motor, 60 Hz													
X: Special configuration													
Approvals of pump													
A: FM/UL													
B: FM													
C: UL													
X: No approval													
Approval of fire pump set													
X: No approval													
Pipe connection													
A: ANSI flange													
D: DIN flange													
Control panel													
A: Mounted on base frame													
F: For floor mounting													
W: For wall mounting													
X: No control panel													
Stuffing box													
B: SNEA													
C: SNFA													
Coupling													
A: Standard (only DNF)													
B: Spacer (only DNF)													
C: Other types													
D: PTO (Power Take-Off shaft)													

The first example shows a fire pump set with a DNF end-suction pump driven by an electric motor. The pump with stuffing box with internal barrier fluid is FM-approved and UL-listed. The diameter of its DIN flange on the discharge side is 80 mm, and the impeller diameter is 260 mm. Pump and motor are connected by a standard coupling. Operation takes place via a controller placed on the common base frame.

6.1.2 Type key for controller

Example	GPA - 380-416 / 50 / 3 / 50
Type	
GPA: Fire pump controller for direct-on-line starting	
GPY: Fire pump controller for star-delta starting	
Supply voltage [V]	
208	
240	
380-416	
480	
600	
Motor power [hp]	
Number of phases	
1	
3	
Frequency	
50: 50 Hz	
60: 60 Hz	

The example shows an controller with microprocessor for direct-on-line starting. The supply voltage is 3 x 380-416 V, 50 Hz. The controller is designed for a 50 hp electric motor.

6.1.3 Type key for stuffing box

Example	S N E A
S: Packing-type stuffing box	
N: Uncooled stuffing box	
E: With internal barrier fluid	
F: With external barrier fluid	
A: PTFE-impregnated fibre packing rings and EPDM O-ring in pump housing	

HSEF and DNF pumps are equipped with an uncooled stuffing box with internal or external barrier fluid. The fibre packing rings are PTFE-impregnated, and the O-ring of the pump housing is made of EPDM.

6.2 Nameplates

All important data of the fire pump set are stated on the fire pump set nameplate (fig. 11), pump nameplate (fig. 12 and 13), the motor nameplate (fig. 14) and controller nameplate (fig. 14).

Nameplate for fire pump set

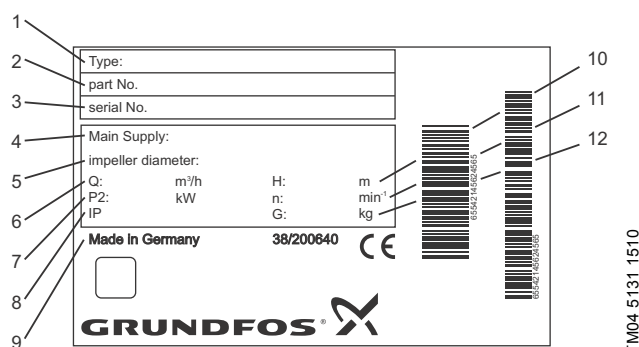


Fig. 11 Nameplate for fire pump set

Pos.	Description
1	Type designation
2	Product number
3	Serial number
4	Type designation of driver
5	Actual impeller diameter
6	Rated flow rate
7	Power output
8	Enclosure class
9	Country of origin
10	Rated head
11	Speed
12	Weight

Nameplate for HSEF pump

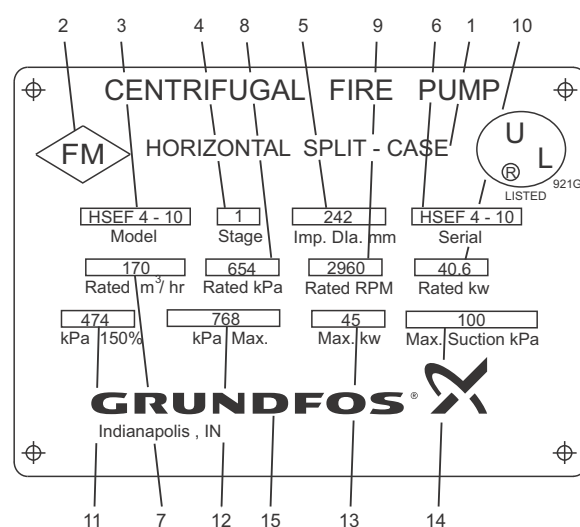


Fig. 12 Example of nameplate for HSEF pump

Pos.	Description
1	Pump type
2	Approvals
3	Type designation
4	Number of stages
5	Actual impeller diameter
6	Serial number
7	Rated flow rate
8	Rated head
9	Rated speed
10	Rated power
11	Head at 150 % flow rate
12	Maximum head
13	Maximum power
14	Maximum suction lift
15	Manufacturer

Nameplate for DNF pump

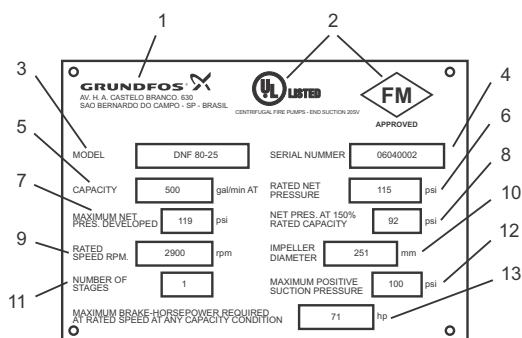


Fig. 13 Example of nameplate for DNF pump

Pos.	Description
1	Manufacturer
2	Approvals
3	Type designation
4	Serial number
5	Rated flow rate
6	Rated head
7	Maximum head
8	Head at 150 % flow rate
9	Rated speed
10	Actual impeller diameter
11	Number of stages
12	Maximum suction lift in PSI
13	Maximum power at rated speed

Nameplate for motor

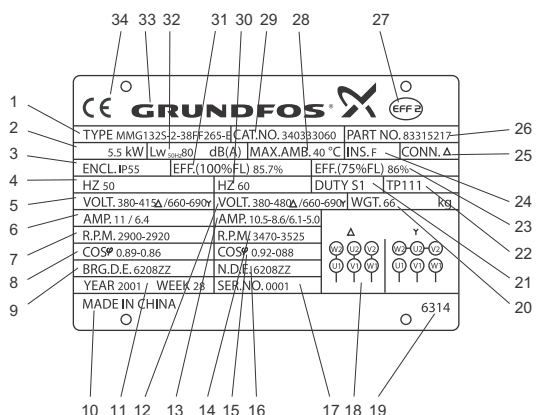


Fig. 14 Example of motor nameplate

Pos.	Description
1	Type designation
2	Motor power
3	Enclosure class
4	50 Hz
5	Voltage supply at 50 Hz
6	Current at 50 Hz
7	Motor speed at 50 Hz
8	cos phi at 50 Hz
9	Bearing, drive end
10	Country of origin
11	Date of production (year - week)
12	Voltage supply at 60 Hz
13	Current at 60 Hz
14	Motor speed at 60 Hz

Pos.	Description
15	cos phi at 60 Hz
16	Bearing, non-drive end
17	Serial number
18	Connection D/Y
19	Nameplate type
20	Weight
21	Duty mode (S1 = continuous operation)
22	TP protection class
23	Efficiency at 75 % of full load
24	Insulation class
25	Connection type
26	Product number
27	Efficiency class
28	Maximum permissible ambient temperature
29	Catalogue number
30	60 Hz
31	Efficiency at 100 % of full load
32	Sound pressure level
33	Manufacturer
34	CE mark

Nameplate for controller

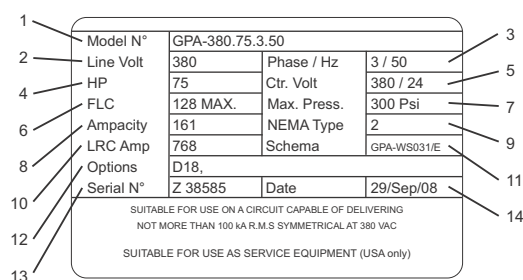


Fig. 15 Example of nameplate for controller

Pos.	Description
1	Type designation
2	Mains voltage
3	Number of phases and frequency
4	Motor power in hp
5	Control voltage
6	Full-load current
7	Maximum pressure of pressure sensor
8	Ampacity (maximum current)
9	Enclosure class (NEMA)
10	Locked-rotor current
11	Drawing number of wiring diagram
12	Options
13	Serial number
14	Date of production

7. Technical data

7.1 Complete fire pump set

Dimensions and weights, see section 19.1.

Sound pressure level

The sound pressure level depends on the motor. It was measured one metre away from the fire pump set. The sound pressure level for the motor types is given in the table below. The motor type is stated on the motor nameplate. See fig. 14, pos. 4.

Motor output [kW]	Sound pressure level [dB(A)]	
	2-pole	4-pole
18.5	78	68
22	82	68
30	84	71
37	84	73
45	84	73
55	85	75
75	86	78
90	86	78
110	89	85
132	89	85
160	92	89
200	92	89
250	95	93
315	95	93



Warning

When the motor is running, the sound pressure level is > 70 dB(A). Hearing protection must therefore be worn when working on or near the motor.

7.2 Pump

See section 19.1 Dimensions and weights.

7.3 Motor

Type:	Grundfos MMG-E, Siemens.
Enclosure class:	IP55.
Insulation class:	F.
Performance:	See section 19.1.
Speed:	See section 19.1.
Weight:	See section 19.1.

7.4 Controller

Type:	Tornatech GPA or GPY.
Control cabinet:	Sheet steel, red.
Enclosure class:	IP42.
Dimensions W x H x D:	24" x 36" x 8".
Weight:	125 to 270 kg.
Supply voltage:	3 x 380 V, 50 or 60 Hz.
Voltage tolerance:	± 10 %.
Maximum current consumption:	Depending on motor size.
EMC-noise emission:	According to EN61000-6-3.
EMC-noise immunity:	According to EN61000-6-2.
Degree of contamination:	2.
Control voltage:	380 V / 24 V.

All optional relays have DPDT contacts rated for 8 A, 250 VAC.

8. Operating conditions

Warning

Do not operate the pump at any pressure, flow rate or liquid temperature other than those for which the pump was originally purchased. Do not pump any other liquid than the one for which the pump was originally purchased without the consent of Grundfos or its authorised representatives. Disregard of this warning can result in pump failure and serious personal injury or death.



8.1 Minimum inlet pressure

At any time, there must be a positive inlet pressure. A minimum inlet pressure of 5 psi is recommended.

Caution

Inadequate inlet pressure will result in cavitation damage to the impeller and casing.

8.2 Maximum inlet pressure

The sum of actual inlet pressure plus pump pressure against a closed valve must always be lower than the maximum permissible operating pressure. See section 8.7 *Maximum operating pressure*. The maximum inlet pressure is 100 psi (approx. 7 bar).

8.3 Minimum flow rate

Caution

Generally, the pump is not allowed to run against a closed valve, as this may cause an increase in temperature or formation of steam in the pump. This may cause damage to the pump.

Only brief periods of operation during start and stop are permissible with a closed discharge valve. Prolonged operation at less than 15 to 20 % of the pump's rated capacity will cause heating of the pumped liquid, impeller erosion, short life of bearings and packing due to stress or vibration. Some pumps may incur shaft damage and wear on stationary parts.

It is therefore advisable to install a bypass pipe to ensure a minimum flow rate of at least 10 % at maximum efficiency. The minimum flow rate helps to dissipate any excessive heat and thus protect the pump against overheating.

Caution

Operation of the pump with flow only through the bypass is not allowed.

How to install a bypass is shown in section 9.6 *Bypass*.

8.4 Pumped liquids

The pumps are suitable for clean and non-aggressive water, not containing solid particles or fibres.

8.5 Liquid temperature

Maximum permissible liquid temperature is 40 °C. The pumps can handle temperature from 0 °C to +120 °C.

8.6 Pump speed

The maximum pump speed is stated on the test report and shown on the nameplate for the fire pump set. See fig. 11.

8.7 Maximum operating pressure

10, 16 or 25 bar.

The maximum operating pressure is stated on the pump nameplate.

8.8 Ambient temperature

Minimum ambient temperature: +4 °C.

Maximum ambient temperature: +40 °C.

8.9 Relative air humidity

The relative air humidity must not be too high, to prevent condensation of moisture in the air, as this can lead to damage of the controller or the motor. If humidity is a problem, install a space heater with a humidistat.

8.10 Effect of ambient temperature and altitude on motor output

If the ambient temperature exceeds +40 °C (fig. 16), or the motor is installed more than 1000 metres above sea level, the motor must not be fully loaded due to the low density and consequently low cooling effect of the air. In these cases, it may be necessary to use a motor with a higher output.

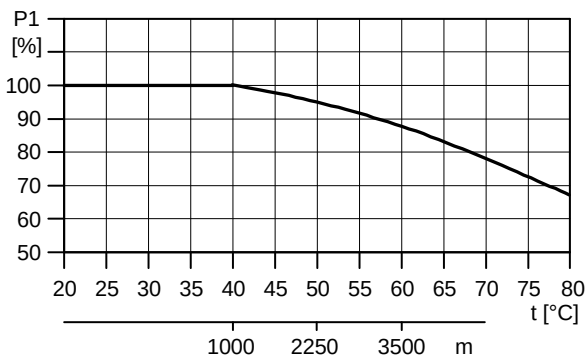


Fig. 16 Relation between motor output, altitude and ambient temperature

9. Installation



Warning

The installation must be carried out in accordance with the following instructions. Non-compliance may result in functional faults which will damage the pump components.

9.1 Installation site

Place the fire pump set in a clean, dry, frost-free and well-ventilated location. It must be secured that unauthorised persons do not have access to the location.

Caution

Ensure an adequate supply of air for cooling the motor and control cabinet.

Allow sufficient clearance around pump and motor to enable inspection, repair and assembly work.

9.2 Foundation

It is advisable to install the fire pump set on a concrete foundation which is heavy enough to provide permanent and rigid support for the entire fire pump set. The foundation must be capable of absorbing any vibration, normal strain or shock. As a rule of thumb, the weight of the concrete foundation should be 1.5 times the weight of the fire pump set. Ordinarily, the proportions used are one part cement to three parts sand and four parts medium aggregate.



Warning

Make sure that weight of the fire pump set and foundation does not exceed the permissible bearing load of the floor.

The concrete foundation must have an absolutely level and even surface. Provide the foundation with anchor bolts to fasten to fasten the fire pump set.

Recommended length and width of the foundation, see fig. 16. Note that the length and width of the foundation must be 200 mm larger than the length and width of the base frame. See dimension x.

The minimum height of the foundation (h_f) can be calculated as follows if the weight of the foundation is assumed to be 1.5 times that of the pump:

$$h_f = \frac{m_{\text{pump}} \times 1.5}{L_f \times B_f \times \delta_{\text{concrete}}}$$

The density (δ) of the foundation is usually taken as 2,200 kg/m³. In installations where low-noise operation is particularly important, we recommend a foundation with a weight of up to five times that of the pump.

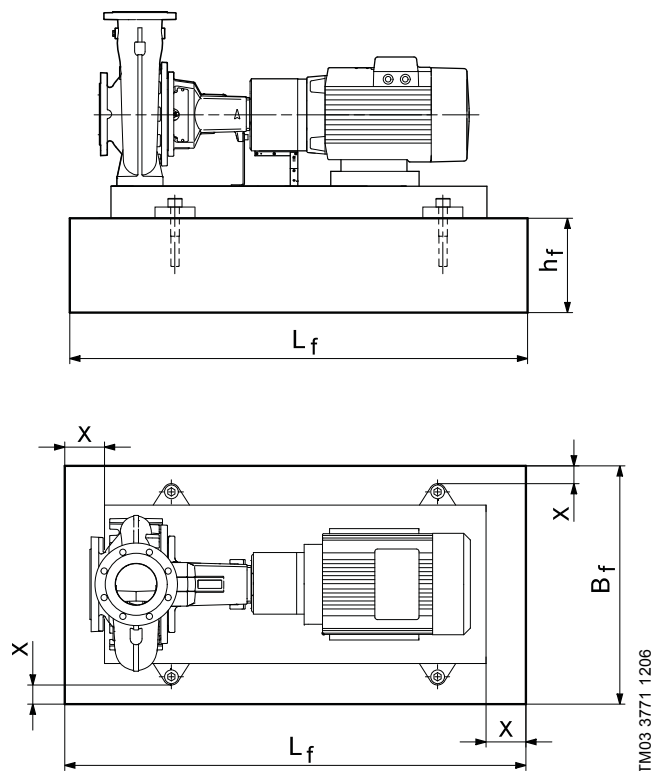


Fig. 17 Dimensions of foundation

9.3 Vibration dampening

To prevent vibrations from being transmitted to the building and pipework, it is advisable to fit expansion joints and vibration dampers. See fig. 18.

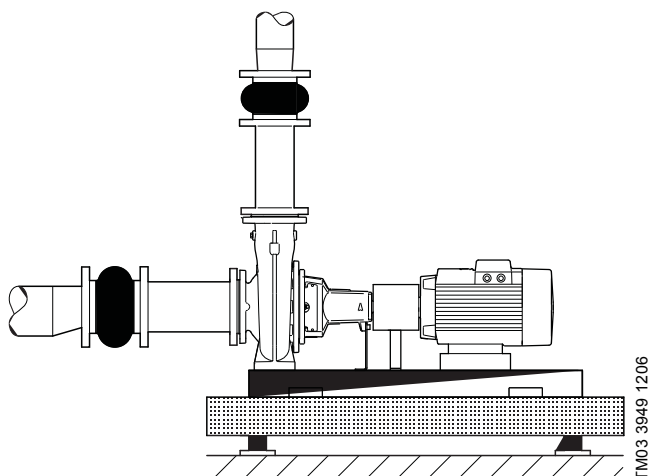


Fig. 18 Vibration dampening

Vibration dampers

To prevent vibrations from being transmitted to the building, it is advisable to isolate the pump foundation from buildings by means of vibration dampers.

To select the correct vibration damper, the following details are required:

- Forces transferred through the damper
- Motor speed
- Desired dampening in % (recommended value: 70 %)

The selection of vibration damper differs from installation to installation. A wrong damper may, in certain cases, increase the vibration level. Vibration dampers should therefore be sized by the supplier.

If the pump is installed on a foundation with vibration dampers, fit expansion joints on both sides of the pump. This prevents the pump from "hanging" from the flanges.

Expansion joints

- Expansion joints provide the following functions:
- Absorption of thermal expansion and contraction of pipework caused by variations in liquid temperature.
- Reduction of mechanical influences in connection with pressure surges in the pipework.
- Isolation of structure-borne noise in the pipework (only rubber bellows expansion joints).

Caution Do not fit expansion joints to make up for inaccuracy in the pipework, e.g. centre displacement or misalignment of flanges.

Fit the expansion joints both on the suction and the discharge sides. The distance from the pump should be at least 1 to 1 1/2 x DN diameter. This prevents turbulence in the joints and ensures optimum suction conditions and minimum pressure drop on the discharge side.

At high water velocities (> 5 m/s), it is advisable to fit larger expansion joints matching the pipework.

9.4 Levelling

Warning

The fire pump set should only be lifted by qualified personnel.

When lifting the entire fire pump set, never use the lifting eyes of the individual components.

Use only suitable lifting equipment in proper condition. See also weight specifications in section 7. Technical data.

During transport of the fire pump set, the cabinet door must be closed and locked.



Place the fire pump set on the foundation or an even floor and anchor it with adequately sized bolts. The entire base frame must be supported uniformly. See figures 19 and 20. If necessary, use shims for horizontal levelling of the base frame.

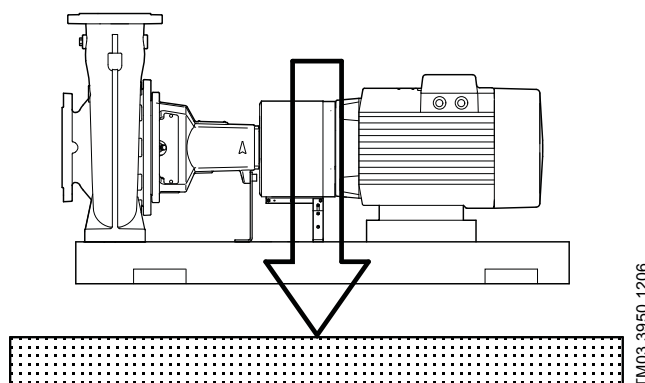


Fig. 19 Correct installation

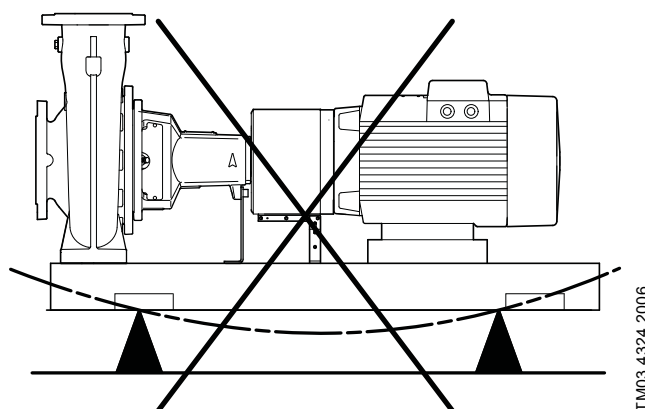


Fig. 20 Incorrect installation

In addition, we recommend fitting vibration dampers under the fire pump set. See section 9.3 Vibration dampening.

9.5 Pipework

Warning

Do not put fingers, hands, arms, etc. into the suction or discharge openings or into any other opening, such as that of the air relief valve.

Do not touch the rotating impeller, as this can cause severe injury.

Do not remove protective covers or packaging until installation.



Caution

The pipework must not stress the pump housing or transfer any forces to the pump housing. Noncompliance may damage the pump.

The suction and discharge piping should be arranged for the simplest, most direct layout. The pipes must be of an adequate size, taking also the pump inlet pressure into consideration. The direction of flow is marked with arrows.

Install the pipes so that air locks are avoided, especially on the suction side of the pump. See fig. 21.

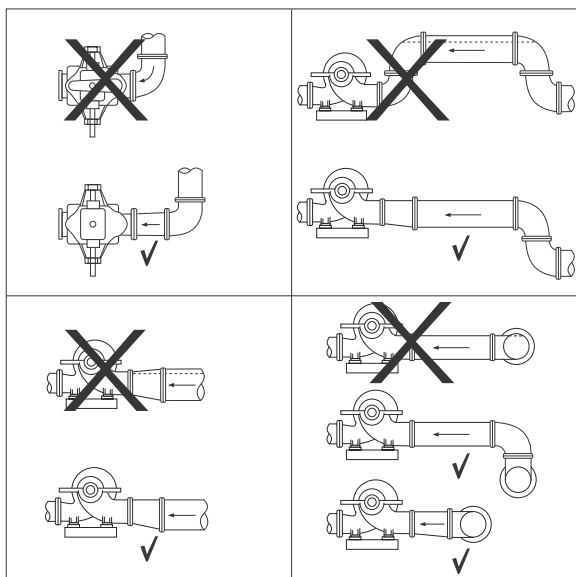


Fig. 21 Correct and incorrect installation of suction pipe

Fit pipe hangers on the wall, ceiling or floor close to the pump on the suction and the discharge side. See fig. 22.

The pipes should lie true against the pump flanges without being stressed. Otherwise, the pump may be damaged.

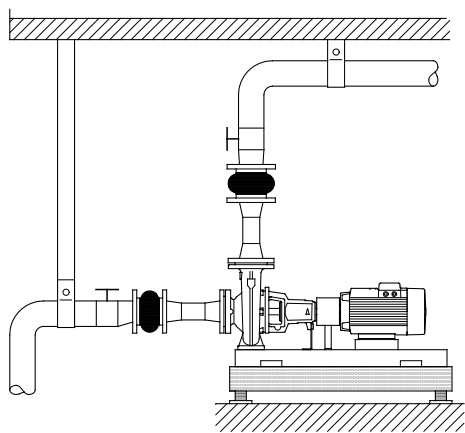


Fig. 22 Fixing points of the pipes

Isolating valves must be installed on the suction and the discharge side of the pump.

9.6 Bypass

A bypass to the suction source, if required, should be made as shown in fig. 23. The distance between bypass pipe and pump discharge flange must be at least ten pipe diameters.

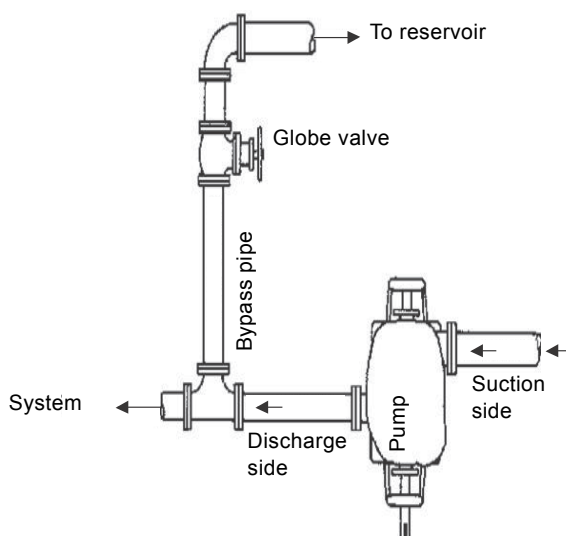


Fig. 23 Possible bypass arrangement

9.7 Connection to the pressure sensor

The threaded connection for the pressure sensor (1/4" NPTF) is on the right side of the controller (fig. 3, pos. 6). Run a hose from the discharge pipe to the pressure sensor. Water pipes must be flushed clean before connecting the pressure sensor.

9.8 Priming tank and test pipe

If the fire pump set is supplied from a storage tank, a pump priming tank must be installed on the suction side in accordance with local regulations.

Install a test pipe running from the discharge pipe to the storage tank. The discharge pipe and the test pipe must be fitted with isolating valves. If the fire pump set is fed directly from the public water supply, the test pipe must have a free outlet. If the test pipe is led back to the reservoir, it must be equipped with a waste cone.

9.9 Alignment



Warning

Before removing the coupling guard, make sure that the fire pump set has been switched off and cannot be accidentally switched on.

Pump and motor are connected by a standard or spacer coupling from FLENDER.

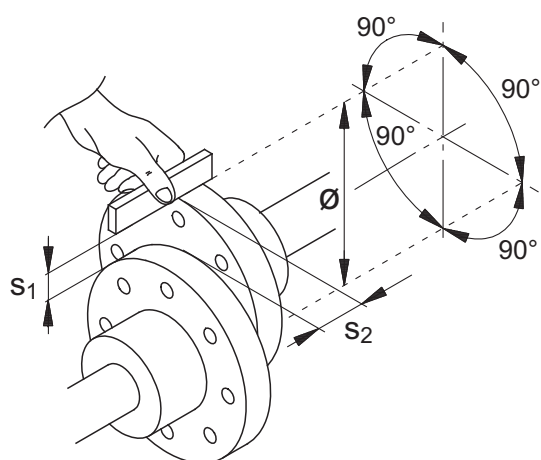
If a fire pump set comes preassembled from factory, the coupling has been accurately aligned by means of shims inserted under the pump and motor.

During transport and installation, the pump-motor alignment may have been affected. It must therefore always be checked when the fire pump set has been installed.

Caution

Careful alignment is important for a long coupling life. This is particularly important with increased motor speed. Follow the installation instructions of the coupling manufacturer.

The coupling used for connecting the pump and motor is shown in fig. 24.

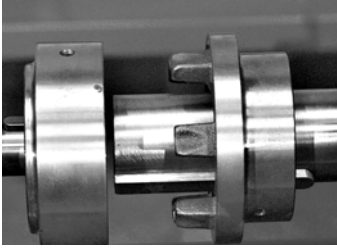
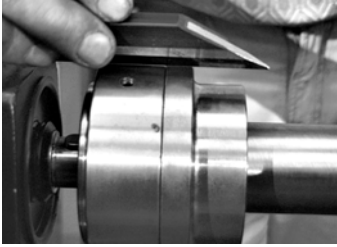
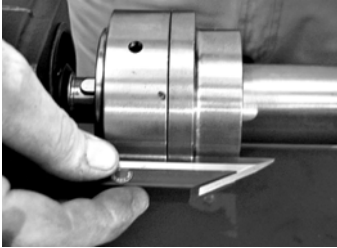
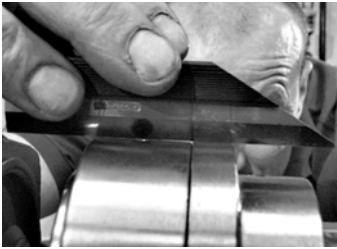


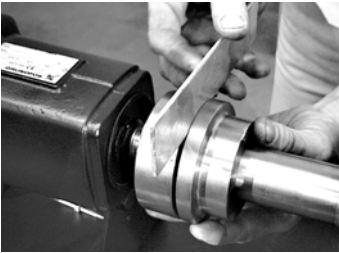
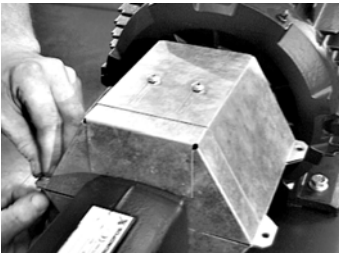
TM03 2999 5005

Fig. 24 Alignment of pump and motor

The values for $\varnothing$ and s_2 can be found in the following table. The value for s_1 is 0.1 mm.

Outside diameter of coupling, $\varnothing$ [mm]	Clearance, s_2 [mm]			
	Standard coupling		Spacer coupling	
	Nominal	Tolerance	Nominal	Tolerance
80	-	-	4	0/-1
95	-	-	4	0/-1
110	-	-	4	0/-1
125	4	0/-1	4	0/-1
140	4	0/-1	4	0/-1
160	4	0/-1	4	0/-1
200	4	0/-1	4	0/-1
225	4	0/-1	4	0/-1
250	4	0/-1	4	0/-1

Step	Action
1	 The distance between the shaft ends must correspond to the value for s_2 stated in the above tables. Displace the shaft keys by 180°. TM01 8880 0800
2	 Check the alignment. TM01 8881 0800
3	 Repeat the alignment check with 90° displacement. See also fig. 24. TM01 8882 0800
4	 Tighten the screws holding the pump and motor to the base frame. TM01 8883 0800
5	 Check the alignment. The tolerance for s_1 is ± 0.1 mm. If the alignment is OK, go to step 10. The picture shows that the pump must be raised. TM01 8884 0800
6	 Cut out the foil in adequate size. TM01 8885 0800

Step	Action
7	 Place the foil where needed. TM01 8886 0800
8	 Retighten the screws. TM01 8887 0800
9	 Check the alignment carefully once more. TM01 8888 0800
10	 The air-gap width s_2 must correspond to the value stated in the above table. The tolerance is ± 0.1 mm. TM01 8889 0800
11	 Tighten the screw in the coupling half. TM01 8890 0800
12	 Fit the coupling guard. TM01 8891 0800

9.10 Separate control cabinet

The controller must be located as close as practical to the motor and within sight of the motor.



Warning

Installation in hazardous locations and potentially explosive environments is not allowed. The controller must be located or protected that it will not be damaged by water escaping from pump or pump connections.

The controller is not designed for outdoor installation. It must not be exposed to direct sunlight and must be installed in a dry, well ventilated but frost-free location.

Caution

Ensure easy accessibility to the controller or its components and adequate air supply for cooling.

Wall-mounted controller

Attach the controller to the structure or wall not less than 305 mm (12 inches) above floor level. Drill holes in the wall according to the drilling template on the rear panel of the control cabinet. Use dowels and screws of adequate size. All four mounting holes must be used.



Warning

When drilling the holes, take care not to damage any cables or water and gas pipes. Ensure a safe installation.

Floor-mounted controller

Attach the controller to the floor using all holes of the mounting feet. The mounting feet provides the required 305 mm (12 inches) clearance for current-carrying parts. A concrete slab is recommended to avoid water accumulation at the controller's feet.

10. Electrical connection



Warning

Check that the supply voltage specified on the nameplate of the control cabinet corresponds to the local conditions.

The electrical connection should be carried out by qualified personnel in accordance with local regulations.

Caution

The dimensional drawings show the area suitable for power and motor connections. No other location must be used. Only watertight hub fittings must be used when entering the cabinet to preserve the NEMA rating of the cabinet.

10.1 Controller on base frame

The electrical connection must be carried out according to the wiring diagram in the control cabinet.

Connection of power supply

- The connection to the power supply must be fitted with a fuse of the correct size in accordance with local regulations and connected to terminals L1-L2 and L3 on the isolating switch. The electronic board is phase-sensitive, and the power conductors must therefore be connected in the correct order of phases. The wiring must typically be sized for minimum 125 % of the full-load current of the pump.

Connection of external sensor signals

- If a pressure switch is installed in the suction pipe, connect it to terminal bar TB12 (fig. 5 or 6, pos. 9). Use one of terminals AL No 1 to 4.
- If a pressure switch is installed in the discharge pipe, connect it to terminal bar TB12 (fig. 5 or 6, pos. 9). Use one of terminals AL No 1 to 4.
- If a pump room thermostat is installed, connect it to terminal bar TB12 (fig. 5 or 6, pos. 9). Use one of terminals AL No 1 to 4.
- If the water reservoir level is to be monitored, connect the float switches for low and for high level to terminal bar TB12 (fig. 5 or 6, pos. 9). Use one of terminals AL No 1 to 4.
- If a vibration sensor is installed in the fire pump set, connect it to terminal bar TB12 (fig. 5 or 6, pos. 9). Use one of terminals AL No 1 to 4.
- If the motor temperature is to be monitored, connect the temperature sensor to terminals 89-90.
- If the status of the flowmeter (option A4) is to be indicated, connect the signal cable of the flowmeter to terminal bar TB12 (fig. 5 or 6, pos. 9). Use one of terminals AL No 1 to 4.

Connection of external control signals

- If the pump is to be started by a remote signal, connect the signal cable to terminal "Remote St NC" (fig. 5 or 6, pos. 9).
- If the pump is to be started by a signal from a deluge valve, connect the valve contact to terminal "Deluge NC" (fig. 5 or 6, pos. 9). Remove jumper J1 to enable the function.
- If the pump is to be started by a signal from a flow switch (NO), connect the signal cable to terminal "Flow switch" (fig. 5 or 6, pos. 9).
- If the fire pump set is to be locked by external equipment (option A6), terminals 3-110 can be used to connect a NO contact when the interlocked equipment is in the same pump room and wiring is mechanically protected. Otherwise, terminal 104-106 must be used when connecting a remote control voltage from the other equipment. Make sure that the control voltage from other equipment and that from control relay CR10 are compatible. For electrical connection, see also the label in the control cabinet.
- If the fire pump sets are divided into zones (options A9 to A11), see the label in the control cabinet.

Connection of external equipment

- If the controller is to lock other equipment (option A7), see the label in the control cabinet.
- If a motor heater is to be connected, see the label in the control cabinet.

All other external sensors must be connected to the terminals shown on the wiring diagram in the control cabinet.

10.2 Wall- and floor-mounted controller

In the case of wall- or floor-mounted controllers, the connections to the motor must also be made besides the connections described in section 10.1 *Controller on base frame*.

Type GPA

The motor must be connected to terminals T1, T2 and T3 on the main contactor (1M). The wiring must typically be sized for minimum 125 % of the full-load current of the pump.

Type GPY

The motor must be connected to terminals T1-T2-T3 on contactor 1M and T4-T5-T6 on contactor 2M. The wiring must typically be sized for minimum 125 % of the full-load current of the pump.

10.3 Connection to standard alarm relays

If operating and alarm messages are to be transferred for instance to a building management system, conductors can be connected to the standard alarm relays (fig. 5 and 6, pos. 9). The alarm relays has one NO and one NC contact.

The table below gives an overview of standard alarm relays.

Relay	Function
RUN	Indicates that the pump is running.
Pow. Av.	Indicates that power supply is okay. If deactivated, there is a phase or power failure.
Ph. Rev.	Indicates phase reversal.
Room Al.	Indicates one of the following conditions: - Overvoltage - Undervoltage - Phase unbalance.
Motor Tr.	Indicates one of the following conditions: - Overcurrent - Undercurrent - Fail to start - Earth fault.
EPSR ready	Indicates that the pressure board (EPSR) is ready and functioning.

10.4 Connection to optional control relays

If more detailed information about operating and alarm conditions is required, conductors can be connected to the optional control relays for transferring operating and alarm messages to for instance a building management system. The alarm relays have one NO and one NC contact. The signal can be transferred to maximum two external controllers. If the NO contact is to be used, connect the signal cable to terminal 14 (24); if the NC contact is to be used, connect the signal cable to terminal 12 (22). The common conductor must be connected to terminal 11 (21). The position of the optional alarm relays (pos. 1) is shown in fig. 5 and 6.

The table below gives an overview of relays.

Option	Relay	Function
C1	-	Extra motor run alarm relay NO: Terminals 93-94 NC: Terminals 95-96
C4	Weekly t.	Weekly test signal
C5	CR18	Alarm, low inlet pressure
C6	CR19	Alarm, low discharge pressure
C7	CR20	Alarm, low ambient temperature
C10	CR21	Alarm, low level in water reservoir
C11	CR22	Alarm, high motor temperature
C12	CR23	Alarm, high vibration
C13	CR24	Alarm, earth fault
C14	CR25	Signal, pump-on-demand
C15	CR26	Alarm, fail to start
C16	CR27	Signal, control voltage okay
C17	CR39	Signal, flowmeter open
C18	CR40	Alarm, high level in water reservoir

11. Start-up



Warning

Start-up may only be carried out by authorised personnel.

The control cabinet must remain closed during start-up. Danger of electric shock!

11.1 Check before start-up

1. Check coupling alignment. See section 9.9 *Alignment*.
2. Check that all bolts are properly tightened.
3. Check that pipework has been installed correctly.
4. Check that all electric connections in and outside the control cabinet.
5. Check that there is enough water in the reservoir.

11.2 Start-up

1. Close the isolating valve on the discharge side of the pump. Open the isolating valve on the suction side of the pump. Open the isolating valve to the test pipe.
2. Turn circuit breaker A1 (fig. 3, pos. 1) to "ON". If the system pressure is lower than the cut-in pressure, the pump will start automatically.
3. Set the cut-in and cut-out pressures. See section 11.3 *Setting of cut-in and cut-out pressures*.
4. If the system pressure is higher than the cut-in pressure, the pump will not start. It must started manually. See section 12.2.1 *Manual operation with buttons [START] and [STOP]*.

Note *If the pump does not start, check the error messages and instruction in the display. Other causes, see section 15. Fault finding.*

5. Slowly open the isolating valve on the discharge side and to the test pipe.
 6. When the pump reaches the operating pressure, open the isolating valve sufficiently to reach the duty point.
 7. Check the direction of rotation. See section 11.4 *Checking the direction of rotation*.
 8. Adjust the stuffing box, if necessary. See section 15.3 *Pump*.
 9. Stop the pump manually. See section 12.2.1 *Manual operation with buttons [START] and [STOP]*.
 10. Simulate a power failure by turning the circuit breaker to "OFF", and check if the circuit connected to terminals "PowAv" is reacting properly.
 11. Simulate phase reversal by pressing the [Alarm Reset] button for more than 10 seconds, and check that the circuit connected to terminals "Ph. Rev." is reacting properly.
- The fire pump set is now operational and in automatic mode.

11.3 Setting of cut-in and cut-out pressures

To enable setting, DIP switch "Lock-Unlock" must be set to "ON". See section 5.1 *Settings via DIP switches*.

It is important to set the cut-out pressure first. It must be lower than the maximum pressure of the pump; otherwise the pump will never stop. The cut-in pressure must be set according to the system pressure.

A quick push on buttons [Cut-in] B3 and [Cut-out] B2 (fig. 4) increases the value by one unit. A long push increases the value by ten units. The value only increases from minimum to maximum. When the maximum value is reached, the value returns to the minimum value.

Note *It is important to set DIP switch "Lock-Unlock" back to "OFF" to prevent unauthorised changes.*

11.4 Checking the direction of rotation

1. Turn circuit breaker A1 to "ON". If the pump does not start, press [START] button A2 (fig. 3, pos. 2).
2. Observe the direction of rotation.
3. If the direction of rotation is not correct, switch off the power supply, and interchange two phases in the terminal box or in the controller.

12. Operation



Warning

The motor will become hot during normal operation. Its surfaces may remain hot for some time after it has been stopped.

The important operating parameters are shown in display D1. See section 4.4.4 *Display D1 (power status)*. If the controller outputs are connected to a building management system, the operation can be remotely monitored.

In case of failure, see sections 4.4.6 *Indicator lights on control panel* and 15. *Fault finding*.

12.1 Automatic operation

Once the fire pump set has been installed and started up according to the instructions, no further preparation is necessary. The fire pump set starts automatically when the sprinkler system is activated, and the pressure switch detects a pressure drop. The fire pump set will also start automatically if it receives a remote start signal or a signal from a deluge valve, when it is time for the weekly test, or in case of a power failure.

When the system pressure reaches the cut-out pressure, the engine will stop if [STOP] button A3 (fig. 3, pos. 3) is pressed, or it will stop automatically if set to automatic stop. See section 4.5.1 *Operating functions*. If not all starting conditions have been eliminated, turn circuit breaker A1 (fig. 3, pos. 1) to "OFF".

If started up due to a false alarm, the pump should not operate for more than 48 hours. If the pump has run for more than 48 hours after a false alarm, always dismantle and inspect it for signs of wear or damage. Replace any defective components.

Note

12.2 Manual operation

The fire pump set can be started manually for a test run, servicing or temporary switching on or off during start-up or in case of a failure.

12.2.1 Manual operation with buttons [START] and [STOP]

Press [START] button A2 (fig. 3, pos. 2) to start the pump regardless of system pressure.

To stop the pump, press [STOP] button A3 (fig. 3, pos. 3). Automatic stop is not possible.

12.2.2 Manual operation with emergency start handle

In case of failure, the fire pump set can also be started with emergency start handle A4 (fig. 3, pos. 4).

To avoid damaging the contactor, it is advisable to start the pump as follows:

1. Switch off the power supply using circuit breaker A1 (fig. 3, pos. 1).
2. Slowly pull the emergency start handle until a resistance is felt. The pump must start before this point.

Note

If the pump does not start before the resistance point, contact Grundfos.

3. Lock the emergency start handle in closed position.
4. Switch on the power supply with the circuit breaker.

The pump always starts at full voltage regardless of system pressure and continues to run.

To stop the pump, switch the emergency start handle back to "OFF", and press [STOP] button A3 (fig. 3, pos. 3).

12.3 Test run

To start a test run, open the isolating valve to the test pipe, and press buttons [Cut-in] B3 and [Cut-out] B2 (fig. 4). The pump runs for ten minutes. Indicator light "Weekly test" is on.

Note

Regular test runs to ensure efficiency should be limited to a maximum of 30 minutes per week.

After ten minutes, the pump stops automatically, and indicator light "Weekly test" is off.

During the test run, the pump can be stopped by pressing [STOP] button A3 (fig. 3, pos. 3). After the test run, close the isolating valve to the test pipe.

12.4 Retrieving and printing data

12.4.1 Retrieving data via the USB port

As standard, the controller has a USB port (fig. 3, pos. 7) for reading of data. The port must be connected to a laptop. A communication software, such as HyperTerminal (provided with the latest Windows version), and "EPSR USB Port Driver" must be installed on the laptop. The driver can be downloaded from www.tornatech.com. The procedure for using HyperTerminal as communication software is described below.

1. Connect the communication port of your laptop to the USB port of the controller.
2. Select from the Windows toolbar: Start > Program > Accessories > Communications > HyperTerminal.
3. If your HyperTerminal software has not been configured, double-click on hypertrm.exe. Otherwise, continue with step 6. **Note:** This procedure may not be available on certain versions of software.
4. In the window "New Connection", enter a connection name, for instance "EPSR", and click [OK].
5. In the next window, set communication parameters as follows:
 - bits per second: 9600
 - data bits: 8
 - parity: none
 - stop bits: 1
 - flow control: hardware.
 Click [OK].
6. On the register "Settings", verify the following selection:
 - Terminal Keys
 - Ctrl+H
 - ANSW
 - VT100
 - 500.
7. Click the button [ASCII Setup] and check that only "Append line feeds to incoming line ends" is marked. Click [OK] to validate in both windows.

If the configuration has been made correctly, this window will be shown when starting HyperTerminal:

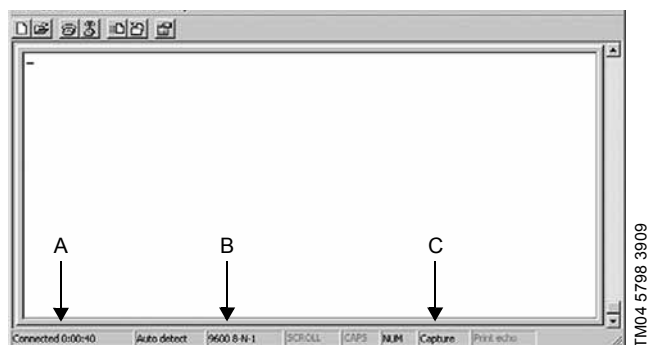


Fig. 25 Start window of HyperTerminal

- A Connected, and timer is counting.
If not, select "Call - call".
- 9600-8-N-1
- B If different, restart a new communication and make port settings ("COM").
If highlighted, data will be stored in the selected file.
Note: If the file is not empty, new data will be stored at the end of the text.
- C If not highlighted, it will not be possible to save downloaded data.
To enable "Capture", click "transfer-capture text - enter file name - start".
8. If your HyperTerminal software has already been configured to communicate with a fire pump controller, select this configuration.
9. Select Transfer > Capture Text.
A window opens so that you can select a file. Enter the name of the file (*.txt) using the browse function.
10. Click [Start] to begin retrieving or downloading data. All new text will be saved in the destination.
11. Click "transfer-capture text-stop" to stop saving incoming text.
New coming text will not be saved.

On the computer keyboard, hit button [1] to get pressure value of both analog inputs, button [2] to download pressure data of the last seven days, button [3] to download a resume of events of the last 15 days, or button [4] to get pressure board (EPSR) settings.

Note

12. To close the session, click on file and select "Exit". Click on [Yes] to disconnect and confirm saving with [Yes].
The downloaded data is saved in the file created in step 4 and selected in step 9. This file is a *.txt file and can be read by programs such as Notepad, Word, Lotus and Excel.
When the file is opened, the *.txt file type must be selected to be shown in the file list.

12.4.2 Printing of data

If the controller is equipped with a printer, it is located behind a plastic cover. The cover protects the printer from humidity and dust. The cover must be lifted when printing. The printer works on demand only. However, in order to prevent a paper drum deformation, the paper drum is activated automatically at regular intervals.

Printing

Press [Print] button B4 (fig. 4) to print a resume of events. When it has been printed, press the button again within 15 seconds.

Paper replacement

The paper must be thermal paper and be correctly oriented. The following operations must be followed to replace the paper: The paper must be thermal paper and be correctly oriented. The following operations must be followed to replace the paper:

1. Unscrew the axle and remove the empty roll.
2. On the printer, open the top cover to get access to the roller.
3. Lift the cover completely.
4. Insert the thermal paper in the slot. The thermal side must be face down.
5. Close the cover.

13. Taking the pump out of operation

Note *The fire pump set must not run at the moment.*

1. Close the isolating valve on the discharge side.
2. Close the isolating valve on the suction side.
3. Turn the circuit breaker A1 (fig. 3 pos. 1) to "OFF". Secure the circuit breaker with a padlock so that the fire pump set cannot be accidentally switched on

During extended periods of inactivity, drain the pump (see figures 26 and 27), and remove and discard the stuffing box packing according to service instructions. Then follow the measures listed in section 3.3 Storage.

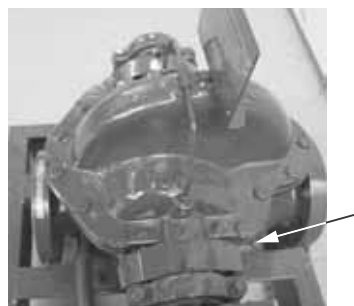


Fig. 26 Drain plug HSEF pump

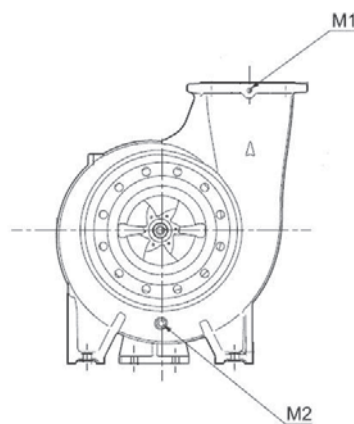


Fig. 27 Drain plug DNF pump (M2)

To restart the fire pump set, remove all protective covers, and fit a new stuffing box packing according to service instructions. Continue as described in section 11. *Start-up.*

14. Maintenance

The operator is responsible for ensuring that all maintenance, inspection and installation work is performed by qualified personnel. A regular maintenance plan will help avoid expensive repairs and contribute to trouble-free, reliable operation. It is recommended to place a maintenance contract with Grundfos.

Note

Warning



Before starting work on the pump, make sure that the power supply has been switched off and cannot be accidentally switched on. Maintenance work must only be carried out by qualified personnel.

Note

More detailed information about maintenance can be found in the service instructions.

14.1 Maintenance intervals

Fire pump set

Visible check:	Every six months.
Tightness of bolts, etc.:	Every six months.
Alignment, motor and pump:	Every six months.

Pump

Stuffing box:	Every six months.
Gaskets:	Every six months.
Lubrication of bearing:	Every six months.
Replacement of bearings:	Every six years.

Motor

Motor cables:	Every six months.
Lubrication of bearings:	Every six months.
Replacement of bearings:	Every six years.
Fan:	Every three years.

Coupling

Coupling guard:	Once a week
Coupling assembly:	Every six months.

Controller

Checking of wire connections:	Every six months.
Visible inspection:	Every six months.

14.2 Fire pump set

A test run should be carried out once a week. See section 12.3 *Test run*. The test should include the following tests:

- Start the pump with [START] button A2. See section 12.2.1 *Manual operation with buttons [START] and [STOP]*.
- Start the pump with emergency start handle A4. See section 12.2.2 *Manual operation with emergency start handle*.
- Simulate a phase reversal by pressing the [Alarm Reset] button for more than 10 seconds, and check indication and remote alarm.
- Start the pump via a remote signal, if possible.
- Start the pump via the flow switch (option A4).

Note

Check that the control panel has returned to normal operation after the test run.

Check the following:

- bolts for tightness and corrosion
- alignment of motor and pump (see section 9.9 *Alignment*)
- any parts for corrosion, damage and wear.

14.3 Pump

Gaskets

Check the gaskets for leakage and wear.

Stuffing box

Check the packing carefully, allowing a slight leakage for lubrication. Never force the packing into a leak-proof position, as this will create excessive friction and cause damage to the packing or shaft sleeve. Maximum packing life can be expected with a leakage of minimum of 40 to 60 drops per minute. A reduction of leakage considerably reduces the life of both the packing and shaft sleeve. If leakage is excessive, tighten the gland bolts evenly, about 1/6 of a turn at a time. Do not be confused if the leakage seems to increase when the packing has been adjusted. The leakage will normally reduce after a period of time as the packing adjusts itself to its new position. It should be kept in mind that it takes time for newly installed packing to "run-in", and that during this initial period, frequent attention and adjustments are necessary. It sometimes takes several days to achieve the desired results.

To set the leakage rate of the stuffing box, remove the protecting guards of the shaft on both sides. This gives access to the setting screw. See service instructions.

Bearing lubrication

If the pump is provided with lubrication nipples, the bearings should be relubricated once a year. Use lithium soap base, worked penetration, meeting NLGI Grade 2 specifications.

Minimum dropping point: 180 °C (355 °F).

Operating temperature range: -23 °C to 116 °C (-10 °F to 240 °F).

The required amount depends on the bearing size. See service instructions.

14.4 Motor

During the weekly test run, check the current drawn by the motor. Lubricate the bearings once a year, and replace the bearings every six years. Clean the fan every three years.

14.5 Coupling

Couplings with elastomeric drive parts do not require lubrication. Check the coupling every six months and the coupling guard for correct position once a week.

14.6 Controller



Warning

Before opening the control cabinet, switch off the power supply by means of the isolating switch. Risk of electric shock!

The following maintenance work must be carried out regularly, at least every six months:

- Check wire connections. All screw connections on the brackets and all frame connections must be checked to ensure a secure fit. Any loose connections must be tightened. Check all cables for visible damage and replace, if necessary. This applies also for the cables at the motor and the signal cables.
- Check that the controller is clean.
- Check the controller for corrosion outside and inside.
- Check for leaks in pressure sensor and pipework.
- Check the door and locks for proper function.
- Check relays, contactors and timers for damage.
- Check controller earthing.
- Check motor contactor power contacts.
- Check tightness of all nuts and screws.
- Check circuit breaker for interlock with the door.

15. Fault finding



Warning

Before carrying out any service of the pump, make sure that the fire pump set cannot accidentally start.

Fault	Cause	Remedy
1. Pump delivers no or too little water.	a) Wrong direction of rotation.	Interchange two phases of the power supply.
	b) Air in suction pipe.	Fill storage container with water. Make sure the suction pipe has been installed according to section 9.5 <i>Pipework</i> .
	c) Pump draws in air due to defective gaskets.	Check the pipeline gaskets and pump housing gaskets, and replace if necessary.
	d) Counter-pressure too high.	Check the system for impurities and blockages.
	e) Inlet pressure too low.	Make sure that the storage tank is filled with enough water, and that the conditions in section 8.1 <i>Minimum inlet pressure</i> are complied with. Open the isolating valve on the suction side completely.
	f) Suction pipe or impeller blocked.	Clean suction pipe and pump.
2. Pump is noisy and/or runs unevenly.	a) Inlet pressure too low (cavitation).	Make sure that the storage tank is filled with enough water and that the conditions in section 8.1 <i>Minimum inlet pressure</i> are complied with. Open the isolating valve on the suction side completely.
	b) Air in suction pipe or pump.	Refill the storage tank with water. Vent pump. Make sure the suction pipe has been installed according to section 9.5 <i>Pipework</i> .
	c) Impeller out of balance.	Clean the impeller.
	d) Inner pump parts worn.	Replace defective parts.
	e) Pump stressed by the pipework.	Install the pump so that it is not stressed. Support the pipes.
	f) Defective bearings.	Replace bearings.
	g) Defective coupling.	Replace coupling.
	h) Foreign bodies in the pump.	Clean the pump and remove foreign bodies.
3. Leakage from pump.	a) Pump stressed by the pipework.	Install the pump so that it is not stressed. Support the pipes.
	b) Pump housing gaskets or pipe gaskets defective.	Replace defective gaskets.
4. Excessive leakage from stuffing box.	a) Stuffing box packing too loose.	Tighten the gland bolts evenly to obtain a leakage of 40 60 drops per minute.
	b) Stuffing box packing worn.	Replace stuffing box packing. See service instructions.
5. Too high pump temperature.	a) Air in the pump.	Fill the storage tank with water.
	b) Inlet pressure too low.	Make sure that the storage tank contains enough water and that all conditions in section 8.1 <i>Minimum inlet pressure</i> are complied with.
	c) Bearings lubricated with too little, too much or unsuitable lubricant.	Replenish, reduce or replace the lubricant.
	d) Pump with bearing seat stressed by pipework.	Install the pump so that it is not stressed. Support the pipes. Check the alignment of the pump, and correct, if necessary. See section 9.9 <i>Alignment</i> .
	e) Axial pressure too high.	Check the relief holes of the impeller and the lock rings on the suction side.

16. Service, spare parts and accessories



Warning

Spare parts and accessories not supplied by Grundfos are not inspected or approved by Grundfos.

The installation and/or use of such products may negatively alter and thus impair the specified properties of the fire pump set.

The usage of non-original spare parts and accessories renders any liability on behalf of Grundfos for resulting damages null and void.

Any malfunctions which cannot be repaired should only be corrected by Grundfos or authorised specialist companies.

Please provide an exact description in the event of a malfunction so that our service technician can prepare and provide the appropriate spare parts.

Please obtain the technical data for the system from the nameplate.

17. Warranty

The warranty is governed by the framework of our general terms of delivery. Liability for any damage resulting from incorrect installation, electrical connection or use, as well as liability for consequential damage, is excluded. The start of the warranty period is to be verified.

18. Further documentation

These operating instructions are to be used in conjunction with the following documentation:

- wiring diagram of controller
- service instructions of the fire pump set.

19. Dimensions, weights and motor data

19.1 Dimensions and weights

Data apply to standard versions.

Fire HSEF, compact unit with controller on base frame, 2-pole

Pump type	Motor type	P2 [kW]	Fire pump set		Suction port [inch]	Discharge port [inch]	Pump		Weight [kg]	Weight [kg]
			Dimensions L x W x H [mm]	Weight [kg]			Height, suction port [mm]	Height, discharge port [mm]		
HSEF 4-10G	MMG 160L-E	18.5	1870 x 701 x 1320	576	5	4	264	264	143	146
	MMG 180M-E	22	1910 x 701 x 1320	600						183
	MMG 200LA-E	30	1980 x 701 x 1320	723						248
	MMG 225M-E	45	2030 x 701 x 1320	792						346
	MMG 250M-E	55	2130 x 701 x 1320	895						462
HSEF 4-12	MMG 250M-E	55	2130 x 701 x 1320	945	5	4	265	233	171	462
	MMG 280S-E	75	2220 x 701 x 1320	1050						525
	MMG 280M-E	90	2271 x 701 x 1320	1096						590
HSEF 5-8	MMG 200LA-E	30	2010 x 701 x 1320	669	6	5	283	283	143	248
	MMG 200LB-E	37	2010 x 701 x 1320	692						270
HSEF 5-11	MMG 225M-E	45	2063 x 701 x 1320	842	6	5	283	283	172	346
	MMG 250M-E	55	2108 x 701 x 1320	946						462
	MMG 280S-E	75	2238 x 701 x 1472	1051						525
	MMG 280M-E	90	2283 x 701 x 1472	1097						590
HSEF 5-12	MMG 280M-E	90	2288 x 701 x 1472	1205	6	5	302	390	257	590
	MMG 315S-E	110	2498 x 853 x 1472	1685						1050
	MMG 315M-E	132	2583 x 853 x 1827	1901						1150
HSEF 6-10	MMG 280S-E	75	2245 x 701 x 1472	1085	8	6	315	334	204	525
	MMG 280M-E	90	2285 x 853 x 1472	1135						590
	MMG 315S-E	110	2475 x 853 x 1472	1611						1050
HSEF 6-12	MMG 315S-E	110	2542 x 853 x 1472	1771	8	6	419	305	340	1050
	MMG 315M-E	132	2798 x 853 x 1827	1992						1150
	MMG 315LA-E	160	2798 x 853 x 1827	2095						1250
	MMG 315LB-E	200	2798 x 853 x 1827	2177						1330

Fire HSEF, flex unit with separate controller, 2-pole

Pump type	Motor type	P2 [kW]	Fire pump set		Suction port [inch]	Discharge port [inch]	Pump		Weight [kg]	Weight [kg]
			Dimensions L x W x H [mm]	Weight [kg]			Height, suction port [mm]	Height, discharge port [mm]		
HSEF 4-10	MMG 160L-E	18.5	1365 x 701 x 678	418	5	4	264	264	143	146
	MMG 180M-E	22	1405 x 701 x 694	442						183
	MMG 200LA-E	30	1475 x 701 x 727	565						248
	MMG 225M-E	45	1525 x 701 x 750	643						346
	MMG 250M-E	55	1625 x 701 x 789	747						462
HSEF 4-12	MMG 250M-E	55	1625 x 701 x 789	787	5	4	265	233	171	462
	MMG 280S-E	75	1665 x 701 x 826	844						525
	MMG 280M-E	90	1716 x 701 x 826	890						590
HSEF 5-8	MMG 200LA-E	30	1502 x 701 x 726	511	6	5	283	283	143	248
	MMG 200LB-E	37	1502 x 701 x 726	534						270
HSEF 5-11	MMG 225M-E	45	1558 x 701 x 776	684	6	5	283	283	172	346
	MMG 250M-E	55	1653 x 701 x 815	788						462
	MMG 280S-E	75	1683 x 701 x 842	845						525
	MMG 280M-E	90	1728 x 701 x 842	891						590
HSEF 5-12	MMG 280M-E	90	1733 x 701 x 891	999	6	5	302	390	257	590
	MMG 315S-E	110	1943 x 853 x 976	1479						1050
	MMG 315M-E	132	2023 x 853 x 976	1591						1150
HSEF 6-10	MMG 280S-E	75	1690 x 701 x 892	879	8	6	315	334	204	525
	MMG 280M-E	90	1730 x 853 x 892	925						590
	MMG 315S-E	110	1920 x 853 x 977	1405						1050
HSEF 6-12	MMG 315S-E	110	2058 x 853 x 1002	1565	8	6	419	305	340	1050
	MMG 315M-E	132	On request	1682						1150
	MMG 315LA-E	160		1785						1250
	MMG 315LB-E	200		1867						1330

Fire DNF, flex unit with separate controller, 2-pole

Pump type	Motor type	P2 [kW]	Fire pump set		Suction port [inch]	Discharge port [inch]	Pump		Weight [kg]	Weight [kg]
			Dimensions L x W x H [mm]	Weight [kg]			Height, suction port [mm]	Height, discharge port [mm]		
DNF 65-20	MMG 160L-E	18.5	1525 x 790 x 678	306	100	65	280	505	71	146
	MMG 180M-E	22	1525 x 790 x 694	343						183
DNF 65-25	MMG 200LB-E	37	1550 x 790 x 727	430	100	65	300	550	69	270
	MMG 225M-E	45	1550 x 790 x 750	506						346
DNF 80-20	MMG 200LA-E	30	1550 x 790 x 727	408	125	80	280	530	72	248
	MMG 200LB-E	37	1550 x 790 x 750	430						270
DNF 80-25	MMG 250M-E	55	1555 x 790 x 789	641	125	80	325	605	82	462
	MMG 280S-E	75	1555 x 790 x 826	735						525

Fire HSEF, flex unit with separate controller, 4-pole

Pump type	Motor type	P2 [kW]	Fire pump set		Suction port [inch]	Discharge port [inch]	Pump		Weight [kg]	Weight [kg]
			Dimensions L x W x H [mm]	Weight [kg]			Height, suction port [mm]	Height, discharge port [mm]		
HSEF 5-14	MMG 180M-E	18.5	1943 x 701 x 1320	707	6	5	302	264	229	180
	MMG 180L-E	22	1983 x 701 x 1320	742						207
	MMG 200L-E	30	2053 x 701 x 1320	809						265
HSEF 6-16	MMG 225M-E	45	2196 x 701 x 1320	966	8	6	299	258	313	350
	MMG 250M-E	55	2276 x 701 x 1320	1081						447
	MMG 280S-E	75	2361 x 701 x 1472	1250						554
HSEF 8-15	MMG 225M-E	45	2205 x 701 x 1320	1059	10	8	328	302	386	350
	MMG 250M-E	55	2280 x 701 x 1320	1195						447
	MMG 280S-E	75	2355 x 701 x 1472	1344						554
HSEF 8-17	MMG 280S-E	75	2399 x 701 x 1472	1400	10	8	353	302	458	554
	MMG 280M-E	90	2449 x 701 x 1472	1447						600
	MMG 315S-E	110	2579 x 853 x 1472	1840						965
	MMG 315M-E	132	2864 x 853 x 1827	2116						1120
	MMG 315LA-E	160	2864 x 853 x 1827	2208						1200
HSEF 8-20	MMG 280M-E	90	2741 x 701 x 1472	1613	10	8	328	227	603	600
	MMG 315S-E	110	2771 x 853 x 1472	1996						965
	MMG 315M-E	132	3056 x 853 x 1827	2275						1120
	MMG 315LA-E	160	3056 x 853 x 1827	2358						1200
	MMG 315LB-E	200	3056 x 853 x 1827	2455						1295
HSEF 10-20	MMG 315LA-E	160	3154 x 853 x 1827	2691	12	10	366	290	907	1200
	MMG 315LB-E	200	3354 x 853 x 1827	2489						1295
	MMG 355M-E	250	3354 x 853 x 1827	3150						1650
HSEF 12-19	MMG 355M-E	250	On request	On request	14	12	241	267	1489	1650
	MMG 355L-E	315								1950
	Siemens 355	355								1900

Fire HSEF, flex unit with separate controller, 4-pole

Pump type	Motor type	P2 [kW]	Fire pump set		Suction port [inch]	Discharge port [inch]	Pump		Weight [kg]	Weight [kg]
			Dimensions L x W x H [mm]	Weight [kg]			Height, suction port [mm]	Height, discharge port [mm]		
HSEF 5-14	MMG 180M-E	18.5	1438 x 701 x 728	549	6	5	302	264	229	180
	MMG 180L-E	22	1478 x 701 x 744	584						207
	MMG 200L-E	30	1548 x 701 x 777	651						265
HSEF 6-16	MMG 225M-E	45	1691 x 701 x 851	808	8	6	299	258	313	350
	MMG 250M-E	55	1771 x 701 x 890	923						447
	MMG 280S-E	75	1806 x 701 x 927	1044						554
HSEF 8-15	MMG 225M-E	45	1700 x 701 x 886	901	10	8	328	302	386	350
	MMG 250M-E	55	1775 x 701 x 915	1037						447
	MMG 280S-E	75	1805 x 701 x 942	1138						554
HSEF 8-17	MMG 280S-E	75	1844 x 701 x 994	1194	10	8	353	302	458	554
	MMG 280M-E	90	1894 x 701 x 994	1241						600
	MMG 315S-E	110	2028 x 853 x 1079	1634						965
	MMG 315M-E	132	2214 x 853 x 1079	1806						1120
	MMG 315LA-E	160	2214 x 853 x 1079	1898						1200
HSEF 8-20	MMG 280M-E	90	2186 x 701 x 993	1407	10	8	328	227	603	600
	MMG 315S-E	110	2216 x 853 x 1078	1790						965
	MMG 315M-E	132	2406 x 853 x 1078	1965						1120
	MMG 315LA-E	160	2406 x 853 x 1078	2048						1200
	MMG 315LB-E	200	2406 x 853 x 1078	2145						1295
HSEF 10-20	MMG 315LA-E	160	2504 x 853 x 1116	2381	12	10	366	290	907	1200
	MMG 315LB-E	200	2704 x 853 x 1284	2479						1295
	MMG 355M-E	250	2704 x 853 x 1284	2840						1650
HSEF 12-19	MMG 355M-E	250	On request	On request	14	12	241	267	1489	1650
	MMG 355L-E	315								1950
	Siemens 355	355								1900

19.2 Motor data

Fire HSEF, 2-pole

Pump type	Motor type	P2 [kW]	Voltage [V]	Speed [min ⁻¹]	I _{1/1} [A]	I _{Start} I _{1/1}
HSEF 4-10G	MMG 160L-E	18.5	3 x 380-415 Δ/660-690 Y	2940	32.5/18.8	6.5
	MMG 180M-E	22		2950	39.5/22.8	7.4
	MMG 200LA-E	30		2960	57.5/33.0	7.0
HSEF 4-10	MMG 225M-E	45	3 x 380-415 Δ/660-690 Y	2980	78.0/45.0	7.4
	MMG 250M-E	55		2960	96.5/55.0	7.9
HSEF 4-12	MMG 250M-E	55	3 x 380-415 Δ/660-690 Y	2960	96.5/55.0	7.9
	MMG 280S-E	75		2970	130/75	6.6
	MMG 280M-E	90		2980	154/89	7.2
HSEF 5-8	MMG 200LA-E	30	3 x 380-415 Δ/660-690 Y	2960	57.5/33.0	7.0
	MMG 200LB-E	37		2960	65.0/37.5	7.6
HSEF 5-11	MMG 225M-E	45	3 x 380-415 Δ/660-690 Y	2980	78.0/45.0	7.4
	MMG 250M-E	55		2960	96.5/55.0	7.9
	MMG 280S-E	75		2970	130/75	6.6
	MMG 280M-E	90		2980	154/89	7.2
HSEF 5-12	MMG 280M-E	90	3 x 380-415 Δ/660-690 Y	2980	154/89	7.2
	MMG 315S-E	110		2980	188/108	7.2
	MMG 315M-E	132		2980	222/128	7.5
HSEF 6-10	MMG 280S-E	75	3 x 380-415 Δ/660-690 Y	2970	130/75	6.6
	MMG 280M-E	90		2980	154/89	7.2
	MMG 315S-E	110		2980	188/108	7.2
HSEF 6-12	MMG 315S-E	110	3 x 380-41 Δ/660-690 Y	2980	188/108	7.2
	MMG 315M-E	132		2980	222/128	7.5
	MMG 315LA-E	160		2980	270/156	6.0
	MMG 315LB-E	200		2980	330/190	5.8

Fire DNF, 2-pole

Pump type	Motor type	P2 [kW]	Voltage [V]	Speed [min ⁻¹]	I _{1/1} [A]	I _{Start} I _{1/1}
DNF 65-20	MMG 160L-E	18.5	3 x 380-415 Δ/660-690 Y	2940	32.5/18.8	6.5
	MMG 180M-E	22		2950	39.5/22.8	7.4
DNF 65-25	MMG 200LB-E	37	3 x 380-415 Δ/660-690 Y	2960	65.0/37.5	7.6
	MMG 225M-E	45		2980	78.0/45.0	7.4
DNF 80-20	MMG 200LA-E	30	3 x 380-415 Δ/660-690 Y	2960	57.5/33.0	7.0
	MMG 200LB-E	37		2960	65.0/37.5	7.6
DNF 80-25	MMG 250M-E	55	3 x 380-415 Δ/660-690 Y	2960	96.5/55.0	7.9
	MMG 280S-E	75		2970	130/75	6.6

Fire HSEF, 4-pole

Pump type	Motor type	P2 [kW]	Voltage [V]	Speed [min ⁻¹]	I _{1/1} [A]	I _{Start} I _{1/1}
HSEF 5-14	MMG 180M-E	18.5	3 x 380-415 Δ/660-690 Y	1465	33.5/19.4	7.6
	MMG 180L-E	22		1465	39.0/22.6	7.8
	MMG 200L-E	30		1470	53.5/31.0	7.5
HSEF 6-16	MMG 225M-E	45	3 x 380-415 Δ/660-690 Y	1480	78/45	7.5
	MMG 250M-E	55		1480	95/55	7.5
	MMG 280S-E	75		1480	128/74	7.4
HSEF 8-15	MMG 225M-E	45	3 x 380-415 Δ/660-690 Y	1480	78/45	7.5
	MMG 250M-E	55		1480	95/55	7.5
	MMG 280S-E	75		1480	128/74	7.4
HSEF 8-17	MMG 280S-E	75	3 x 380-415 Δ/660-690 Y	1480	128/74	7.4
	MMG 280M-E	90		1480	150/86.5	7.5
	MMG 315S-E	110		1490	192/110	7.3
	MMG 315M-E	132		1490	226/130	6.7
	MMG 315LA-E	160		1490	270/156	6.7
HSEF 8-20	MMG 280M-E	90	3 x 380-415 Δ/660-690 Y	1480	150/86.5	7.5
	MMG 315S-E	110		1490	192/110	7.3
	MMG 315M-E	132		1490	226/130	6.7
	MMG 315LA-E	160		1490	270/156	6.7
	MMG 315LB-E	200		1490	340/196	5.5
HSEF 10-20	MMG 315LA-E	160	3 x 380-415 Δ/660-690 Y	1490	270/156	6.7
	MMG 315LB-E	200		1490	340/196	5.5
	MMG 355M-E	250		1490	410/236	6.4
HSEF 12-19	MMG 355M-E	250	3 x 380-415 Δ/660-690 Y	1490	410/236	6.4
	MMG 355L-E	315		1490	525/300	6.8
	Siemens 355	355		1490	610/355	6.5

20. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

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